High-Performance Big Data Analytics Using R

NYC Yellow Taxi Trip Records, 2024

Laboratory Assignment 8 – R Programming

Author: Swaroop Naik (@Swaroop192005)

Contents

1. Problem Statement
2. Dataset
3. Task 1 – Data Acquisition and Preparation
4. Task 2 – Transportation Data Analytics
5. Task 3 – Functional vs Vectorized vs data.table Programming
6. Task 4 – Parallel Computing (foreach + doParallel)
7. Task 5 – Performance Benchmarking and Optimization
8. Task 6 – Data Visualization (ggplot2 + leaflet)
9. Observations
10. Task 7 – Critical Analysis and Recommendation
11. Conclusion

1. Problem Statement

The New York City Taxi and Limousine Commission (NYC TLC) publishes millions of taxi-trip records every month. Each record contains pickup/drop-off timestamps and locations, passenger count, trip distance, fare, tip, tolls, payment type and the total amount charged. Processing data at this scale with conventional row-by-row R code is slow and memory hungry.

Objective: build a high-performance and scalable analytics workflow in R that

- prepares and joins ~20 million trip records with the taxi-zone lookup table using **data.table**,
- identifies travel patterns, high-demand periods, high-revenue routes, fare/distance characteristics and payment preferences,
- compares **functional programming** (`apply()` , `lapply()` , `purrr::map()`) with **vectorized** and **data.table** implementations,
- compares **sequential** and **parallel** execution (`foreach` + `doParallel`) on month-wise partitions,
- benchmarks every approach with `microbenchmark` / `system.time()` and memory measurements, and

- critically evaluates the approaches for execution time, memory, scalability and readability.

2. Dataset

Item	Description
Source	NYC TLC Trip Record Data – https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page
Trip files	Yellow Taxi Trip Records 2024: **January–June** locally (6 files, ≈ 20 M trips) or **January–March** on Colab (3 files, ≈ 9 M trips), set by `MONTHS`
Lookup table	Taxi Zone Lookup Table (`taxi_zone_lookup.csv` , 265 zones) + Taxi Zone shapefile (for the map)

Main attributes used:

Column	Meaning
`tpep_pickup_datetime` , `tpep_dropoff_datetime`	Pickup / drop-off date and time
`PULocationID` , `DOLocationID`	TLC taxi-zone IDs of pickup / drop-off
`passenger_count`	Number of passengers
`trip_distance`	Trip distance in miles
`payment_type`	1 = Credit card, 2 = Cash, 3 = No charge, 4 = Dispute, 0 = Flex fare / unknown
`fare_amount` , `tip_amount` , `tolls_amount` , `total_amount`	Monetary amounts in USD

Six months were chosen because they give a sufficiently large dataset (≈ 3 million rows per month) so that the performance differences between conventional, optimized, sequential and parallel implementations are clearly visible, while still fitting in the memory of a normal laptop. Change `MONTHS` in the next cell to use fewer months on a machine with less than 8 GB RAM.

2.1 Environment setup

Install (if needed) and load the packages. `TZ = "UTC"` is set so that the timestamps, which the TLC stores as New York local time without a time zone, are kept exactly as recorded.

The notebook runs on a local machine and on **Google Colab with an R runtime** (Runtime → Change runtime type → R). On Colab it automatically:

- analyses **three months instead of six**, because a Colab session has about 12 GB of RAM while six months need roughly 11 GB at peak — edit `MONTHS` below to change this;
- installs the system libraries that the `sf` mapping package needs;

- uses **at least two worker cores**, since a Colab session has only two.

The optional map section (`sf` + `leaflet`) is skipped automatically if those packages cannot be installed, so the rest of the notebook always runs.

In [61]:

```
ON_COLAB <- dir.exists("/content")                          # Google Colab R
runtime
CRAN     <- "https://cloud.r-project.org"
have     <- function(p) requireNamespace(p, quietly = TRUE)
need     <- function(pkgs) { p <- pkgs[!vapply(pkgs, have, logical(1))]
                             if (length(p)) install.packages(p, repos =
CRAN, quiet = TRUE) }

# 1. core packages used by every section
need(c("data.table", "ggplot2", "foreach", "doParallel", "microbenchmark",
       "purrr", "lubridate", "scales", "hexbin"))

# 2. a Parquet reader: arrow preferred, nanoparquet as a lightweight
fallback
need("arrow")
if (!have("arrow")) need("nanoparquet")
HAVE_ARROW <- have("arrow")
stopifnot("need either arrow or nanoparquet to read Parquet files" =
HAVE_ARROW || have("nanoparquet"))

# 3. optional mapping packages (advanced task in Task 6); sf needs system
libraries on Colab
if (ON_COLAB && !have("sf")) system("apt-get install -y -qq libgdal-dev
libgeos-dev libproj-dev libudunits2-dev")
try(need(c("sf", "leaflet", "htmlwidgets")), silent = TRUE)
HAVE_MAPS <- all(vapply(c("sf", "leaflet", "htmlwidgets"), have,
logical(1)))

Sys.setenv(TZ = "UTC")
Sys.setenv(ARROW_DEFAULT_MEMORY_POOL = "system")   # let arrow return
memory to the OS after import
suppressPackageStartupMessages({
  library(data.table)
  if (HAVE_ARROW) library(arrow)
  library(ggplot2)
  library(foreach)
  library(doParallel)
  library(microbenchmark)
  library(purrr)
  library(lubridate)
  library(scales)
})

set.seed(2024)
options(repr.plot.width = 12, repr.plot.height = 6, scipen = 999, width =
140)

MONTHS   <- if (ON_COLAB) sprintf("2024-%02d", 1:3) else sprintf("2024-
%02d", 1:6)
# everything below derives the analysis period from MONTHS, so changing
MONTHS is enough
MONTH_NUMS   <- as.integer(substr(MONTHS, 6, 7))
PERIOD_START <- as.POSIXct(paste0(MONTHS[1], "-01 00:00:00"), tz = "UTC")
PERIOD_END   <-
```

```
as.POSIXct(paste0(seq(as.Date(paste0(MONTHS[length(MONTHS)], "-01")),
                                        by = "month", length.out = 2)[2], "
00:00:00"), tz = "UTC")
PERIOD_LABEL <- sprintf("%s-%s 2024", month.abb[MONTH_NUMS[1]],
month.abb[MONTH_NUMS[length(MONTH_NUMS)]])
DATA_DIR <- if (ON_COLAB) "/content/data" else "data"
OUT_DIR  <- if (ON_COLAB) "/content/outputs" else "outputs"
FIG_DIR  <- file.path(OUT_DIR, "figures")
for (d in c(DATA_DIR, FIG_DIR)) dir.create(d, showWarnings = FALSE,
recursive = TRUE)

# keep one core free for the OS, but always use at least two workers so
that the
# sequential-vs-parallel comparison is meaningful on a 2-core machine such
as Colab
N_CORES <- max(2L, parallel::detectCores() - 1L)

cat("R version          :", R.version.string, "\n")
cat("data.table version :", as.character(packageVersion("data.table")),
"\n")
cat("Running on Colab   :", ON_COLAB, "| period analysed:", PERIOD_LABEL,
"\n")
cat("Parquet reader     :", if (HAVE_ARROW) "arrow" else "nanoparquet",
    "| mapping packages available:", HAVE_MAPS, "\n")
cat("Logical cores      :", parallel::detectCores(), "| cores used for
parallel work:", N_CORES, "\n")
cat("data.table threads :", getDTthreads(), "\n")
```

```
R version          : R version 4.6.1 (2026-06-24)
data.table version : 1.18.6.1
Running on Colab   : TRUE | period analysed: Jan-Mar 2024
Parquet reader     : arrow | mapping packages available: TRUE
Logical cores      : 2 | cores used for parallel work: 2
data.table threads : 1
```

2.2 Helper functions

- `peak_mem_mb()` – peak extra memory (MB) used by R while evaluating an expression, measured with `gc()`.
- `save_plot()` – saves a ggplot to `outputs/figures/` and returns it so it is also shown in the notebook.

In [62]:
```
peak_mem_mb <- function(expr) {
  invisible(gc(reset = TRUE))
  before <- sum(gc()[, 2])                  # MB in use now
  force(expr)
  peak <- sum(gc()[, 6])                    # max MB used since reset
  round(peak - before, 1)
}

save_plot <- function(p, name, width = 11, height = 6) {
  ggsave(file.path(FIG_DIR, paste0(name, ".png")), p, width = width,
height = height, dpi = 120, bg = "white")
  p
}

theme_set(theme_minimal(base_size = 13) +
          theme(plot.title = element_text(face = "bold"), legend.position
= "bottom"))
```

2.3 Download the data

Files are downloaded once from the official TLC CloudFront location and re-used afterwards (the `data/` folder is not committed to Git).

In [63]:
```
options(timeout = 3600)
base_url   <- "https://d37ci6vzurychx.cloudfront.net"
trip_files <- file.path(DATA_DIR, sprintf("yellow_tripdata_%s.parquet",
MONTHS))

for (i in seq_along(MONTHS)) {
  if (!file.exists(trip_files[i])) {
    download.file(sprintf("%s/trip-data/yellow_tripdata_%s.parquet",
base_url, MONTHS[i]),
                  trip_files[i], mode = "wb")
  }
}
zone_file <- file.path(DATA_DIR, "taxi_zone_lookup.csv")
if (!file.exists(zone_file)) download.file(paste0(base_url,
"/misc/taxi_zone_lookup.csv"), zone_file, mode = "wb")
shape_zip <- file.path(DATA_DIR, "taxi_zones.zip")
if (!file.exists(shape_zip)) download.file(paste0(base_url,
"/misc/taxi_zones.zip"), shape_zip, mode = "wb")

data.table(file = basename(c(trip_files, zone_file, shape_zip)),
           size_MB = round(file.size(c(trip_files, zone_file, shape_zip))
/ 2^20, 1))
```

A data.table: 5 × 2

file	size_MB
<chr>	<dbl>
yellow_tripdata_2024-01.parquet	47.6
yellow_tripdata_2024-02.parquet	48.0
yellow_tripdata_2024-03.parquet	57.3
taxi_zone_lookup.csv	0.0
taxi_zones.zip	1.0

---

3. Task 1 – Data Acquisition and Preparation

3.1 Import the trip records

Each monthly Parquet file is read with `arrow` (or `nanoparquet` when arrow is unavailable) and the monthly tables are stacked with `data.table::rbindlist()`. Iterating over **files** with `lapply()` is a good use of functional programming: the loop body runs only a handful of times, so the per-call overhead is irrelevant — unlike looping over 20 million rows, which Task 3 measures.

In [64]:
```
read_trip_file <- function(f) {
  if (HAVE_ARROW) as.data.frame(arrow::read_parquet(f)) else
as.data.frame(nanoparquet::read_parquet(f))
}
t_import <- system.time({
  monthly_tables <- lapply(trip_files, read_trip_file)   # functional
iteration over the monthly files
  trips <- rbindlist(monthly_tables, fill = TRUE)        # data.table:
stack them into one table
  rm(monthly_tables); invisible(gc())
})
cat(sprintf("Imported %s rows x %d columns in %.1f s\n",
            format(nrow(trips), big.mark = ","), ncol(trips),
t_import[["elapsed"]]))
raw_size <- object.size(trips)
cat("Memory used by raw table:", format(raw_size, units = "auto"), "\n")
```

```
Imported 9,554,778 rows x 19 columns in 9.4 s
Memory used by raw table: 1.1 Gb
```

3.2 Inspect structure, dimensions, data types and summary statistics

In [65]:
```
dim(trips)
str(trips)
```

9554778 · 19

```
Classes ‘data.table’ and 'data.frame':  9554778 obs. of  19 variables:
 $ VendorID             : int  2 1 1 1 1 1 2 1 2 2 ...
 $ tpep_pickup_datetime : POSIXct, format: "2024-01-01 00:57:55" "2024-01-01 00:03:00" "2024-01-01
00:17:06" "2024-01-01 00:36:38" ...
 $ tpep_dropoff_datetime: POSIXct, format: "2024-01-01 01:17:43" "2024-01-01 00:09:36" "2024-01-01
00:35:01" "2024-01-01 00:44:56" ...
 $ passenger_count      : int  1 1 1 1 1 1 2 0 1 1 ...
 $ trip_distance        : num  1.72 1.8 4.7 1.4 0.8 ...
 $ RatecodeID           : int  1 1 1 1 1 1 1 1 1 1 ...
 $ store_and_fwd_flag   : chr  "N" "N" "N" "N" ...
 $ PULocationID         : int  186 140 236 79 211 148 138 246 161 113 ...
 $ DOLocationID         : int  79 236 79 211 148 141 181 231 261 113 ...
 $ payment_type         : int  2 1 1 1 1 1 1 2 2 2 ...
 $ fare_amount          : num  17.7 10 23.3 10 7.9 29.6 45.7 25.4 31 3 ...
 $ extra                : num  1 3.5 3.5 3.5 3.5 3.5 6 3.5 1 1 ...
 $ mta_tax              : num  0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 0.5 ...
 $ tip_amount           : num  0 3.75 3 2 3.2 6.9 10 0 0 0 ...
 $ tolls_amount         : num  0 0 0 0 0 0 0 0 0 0 ...
 $ improvement_surcharge: num  1 1 1 1 1 1 1 1 1 1 ...
 $ total_amount         : num  22.7 18.8 31.3 17 16.1 ...
 $ congestion_surcharge : num  2.5 2.5 2.5 2.5 2.5 2.5 0 2.5 2.5 2.5 ...
 $ Airport_fee          : num  0 0 0 0 0 0 1.75 0 0 0 ...
 - attr(*, ".internal.selfref")=<pointer: 0x5aa76918b510>
```

In [66]:
```
num_cols <- names(trips)[vapply(trips, is.numeric, logical(1))]
stats <- rbindlist(lapply(num_cols, function(cn) {
  x <- trips[[cn]]
  data.table(column = cn, min = min(x, na.rm = TRUE), mean = mean(x, na.rm
= TRUE),
             median = median(x, na.rm = TRUE), max = max(x, na.rm = TRUE),
missing = sum(is.na(x)))
}))
stats[, (2:5) := lapply(.SD, round, 2), .SDcols = 2:5]
stats
cat("Pickup time range:", format(range(trips$tpep_pickup_datetime)), "\n")
```

A data.table: 16 × 6

column	min	mean	median	max	missing
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<int>
VendorID	1.00	1.76	2.00	6.00	0
passenger_count	0.00	1.33	1.00	9.00	751962
trip_distance	0.00	4.04	1.70	312722.30	0
RatecodeID	1.00	2.15	1.00	99.00	751962
PULocationID	1.00	165.20	162.00	265.00	0
DOLocationID	1.00	164.19	162.00	265.00	0
payment_type	0.00	1.12	1.00	4.00	0
fare_amount	-999.00	18.33	12.80	9792.00	0
extra	-7.50	1.42	1.00	14.25	0
mta_tax	-0.50	0.48	0.50	35.84	0
tip_amount	-300.00	3.27	2.65	999.99	0
tolls_amount	-84.30	0.53	0.00	163.00	0
improvement_surcharge	-1.00	0.98	1.00	1.00	0
total_amount	-1000.00	26.87	20.20	9792.00	0
congestion_surcharge	-2.50	2.26	2.50	2.50	751962
Airport_fee	-1.75	0.14	0.00	1.75	751962

```
Pickup time range: 2002-12-31 22:17:10 2024-04-01 00:34:55
```

3.3 Missing, duplicate, inconsistent and invalid records

First we count missing values per column.

In [67]:
```
na_counts <- trips[, lapply(.SD, function(x) sum(is.na(x)))]
na_counts <- melt(na_counts, measure.vars = names(na_counts),
variable.name = "column", value.name = "missing")
na_counts[, pct := round(100 * missing / nrow(trips), 2)]
na_counts[missing > 0]
# Rows with missing passenger_count also miss RatecodeID /
store_and_fwd_flag and have payment_type 0
trips[is.na(passenger_count), .N, by = payment_type]
```

A data.table: 5 × 3

column	missing	pct
<fct>	<int>	<dbl>
passenger_count	751962	7.87
RatecodeID	751962	7.87
store_and_fwd_flag	751962	7.87
congestion_surcharge	751962	7.87
Airport_fee	751962	7.87

A data.table: 1 × 2

payment_type	N
<int>	<int>
0	751962

Cleaning rules (each step is logged):

Problem	Rule	Action
Duplicates	identical vendor, timestamps, locations, distance and amount	remove
Out-of-period records	pickup outside 1 Jan – 30 Jun 2024 (wrong meter clocks)	remove
Missing `passenger_count` / `RatecodeID`	≈ 10 % of rows (flex-fare / app trips)	**impute** passenger count with the median (1), RatecodeID with 99 = unknown
Zero passengers	inconsistent with a paid trip	impute with median (1)
Non-positive or extreme amounts	fare ≤ 0 or > 500 USD, total ≤ 0, negative tip/tolls	remove (refunds, voids, typos)
Invalid distance	distance ≤ 0 or > 100 miles	remove
Invalid duration	drop-off not after pickup or trip longer than 3 hours	remove
Impossible speed	average speed > 80 mph	remove
Invalid zone IDs	outside 1–265	remove

In [68]:
```
# Memory-efficient cleaning: every rule only updates one logical "keep"
vector (80 MB for 20M rows)
# and the large table is subset ONCE at the end, instead of being copied
after every rule.
n_raw     <- nrow(trips)
keep      <- rep(TRUE, n_raw)
clean_log <- data.table(step = "Raw records", removed = 0L)
apply_rule <- function(step, ok) {
  ok[is.na(ok)] <- FALSE
  clean_log <<- rbind(clean_log, data.table(step = step, removed =
```

```
sum(keep & !ok)))
  keep <<- keep & ok
}

# trip duration in minutes (vectorized, added by reference)
trips[, trip_duration_min := as.numeric(tpep_dropoff_datetime -
tpep_pickup_datetime, units = "mins")]

apply_rule("Remove duplicate trips", !duplicated(trips, by = c("VendorID",
"tpep_pickup_datetime", "tpep_dropoff_datetime",

"PULocationID", "DOLocationID", "trip_distance", "total_amount")))
# a monthly file holds a few trips whose meter clock spills into the next
month; bounding the period by the
# months actually loaded stops those few rows forming a spurious extra
partition in Task 4
apply_rule(paste("Remove pickups outside", PERIOD_LABEL),
           trips[, tpep_pickup_datetime >= PERIOD_START &
tpep_pickup_datetime < PERIOD_END])
apply_rule("Remove invalid fare / total / tip / tolls",
           trips[, fare_amount > 0 & fare_amount <= 500 & total_amount > 0
& tip_amount >= 0 & tolls_amount >= 0])
apply_rule("Remove invalid trip distance", trips[, trip_distance > 0 &
trip_distance <= 100])
apply_rule("Remove invalid trip duration", trips[, trip_duration_min > 0 &
trip_duration_min <= 180])
apply_rule("Remove speed > 80 mph", trips[, trip_distance /
(trip_duration_min / 60) <= 80])
apply_rule("Remove invalid zone IDs", trips[, PULocationID %between% c(1L,
265L) & DOLocationID %between% c(1L, 265L)])

trips <- trips[keep]          # the only copy of the large table
rm(keep); invisible(gc())

# impute missing / zero passenger count and missing rate code (update by
reference)
n_pass_fixed <- trips[is.na(passenger_count) | passenger_count == 0, .N]
trips[is.na(passenger_count) | passenger_count == 0, passenger_count :=
1L]
trips[is.na(RatecodeID), RatecodeID := 99L]
cat("Passenger count imputed for", format(n_pass_fixed, big.mark = ","),
"rows\n")

clean_log[, rows_left := n_raw - cumsum(removed)]
clean_log[, pct_of_raw := round(100 * rows_left / n_raw, 2)]
clean_log
```

```
Passenger count imputed for 730,807 rows
```

A data.table: 8 × 4

step	removed	rows_left	pct_of_raw
<chr>	<int>	<int>	<dbl>
Raw records	0	9554778	100.00
Remove duplicate trips	1	9554777	100.00
Remove pickups outside Jan-Mar 2024	21	9554756	100.00
Remove invalid fare / total / tip / tolls	139712	9415044	98.54
Remove invalid trip distance	204103	9210941	96.40
Remove invalid trip duration	6293	9204648	96.34
Remove speed > 80 mph	2965	9201683	96.30
Remove invalid zone IDs	0	9201683	96.30

3.4 Convert attributes into appropriate data types

Integer codes become labelled **factors** (stored as integers + labels – memory efficient and readable).

In [69]:
```
payment_labels <- c("0" = "Flex/Unknown", "1" = "Credit card", "2" =
"Cash",
                    "3" = "No charge", "4" = "Dispute", "5" = "Unknown",
"6" = "Voided")
rate_labels <- c("1" = "Standard", "2" = "JFK", "3" = "Newark", "4" =
"Nassau/Westchester",
                 "5" = "Negotiated", "6" = "Group ride", "99" = "Unknown")

trips[, payment_type := factor(payment_labels[as.character(payment_type)],
                               levels = c("Credit card", "Cash",
"Flex/Unknown", "Dispute", "No charge", "Unknown", "Voided"))]
trips[, RatecodeID := factor(rate_labels[as.character(RatecodeID)], levels
= unique(rate_labels))]
trips[, passenger_count := as.integer(passenger_count)]
trips[, `:=`(PULocationID = as.integer(PULocationID), DOLocationID =
as.integer(DOLocationID))]
sapply(trips[, .(payment_type, RatecodeID, passenger_count, PULocationID,
trip_distance)], class)
```

payment_type: 'factor' **RatecodeID:** 'factor' **passenger_count:** 'integer' **PULocationID:** 'integer' **trip_distance:** 'numeric'

3.5 Extract temporal attributes

Hour, day of month, day of week (Monday first), month, date and a weekend flag. data.table's `hour()`, `mday()`, `wday()` and `month()` are fast vectorized C implementations.

In [70]:
```
wday_labels <- c("Mon", "Tue", "Wed", "Thu", "Fri", "Sat", "Sun")
t_time <- system.time({
  trips[, `:=`(
    pickup_date  = as.IDate(tpep_pickup_datetime),
```

```
    pickup_hour  = hour(tpep_pickup_datetime),
    pickup_day   = mday(tpep_pickup_datetime),
    pickup_wday  = factor((wday(tpep_pickup_datetime) + 5L) %% 7L + 1L,
levels = 1:7, labels = wday_labels),
    month        = factor(month(tpep_pickup_datetime), levels =
MONTH_NUMS, labels = month.abb[MONTH_NUMS])
  )]
  trips[, is_weekend := pickup_wday %in% c("Sat", "Sun")]
  trips[, time_period := cut(pickup_hour, breaks = c(-1, 5, 9, 15, 19,
23),
                             labels = c("Early morning (0-5)", "Morning
rush (6-9)", "Midday (10-15)",
                                        "Evening rush (16-19)", "Night (20-
23)"))]
})
cat("Temporal features created in", t_time[["elapsed"]], "s\n")
head(trips[, .(tpep_pickup_datetime, pickup_date, pickup_hour, pickup_day,
pickup_wday, month, is_weekend, time_period)])
```

```
Temporal features created in 24.456 s
```

A data.table: 6 × 8

tpep_pickup_datetime	pickup_date	pickup_hour	pickup_day	pickup_wday	month	is_w
<dttm>	<IDate>	<int>	<int>	<fct>	<fct>	
2024-01-01 00:57:55	2024-01-01	0	1	Mon	Jan	
2024-01-01 00:03:00	2024-01-01	0	1	Mon	Jan	
2024-01-01 00:17:06	2024-01-01	0	1	Mon	Jan	
2024-01-01 00:36:38	2024-01-01	0	1	Mon	Jan	
2024-01-01 00:46:51	2024-01-01	0	1	Mon	Jan	
2024-01-01 00:54:08	2024-01-01	0	1	Mon	Jan	

3.6 Remove unnecessary attributes to reduce memory

Surcharges and technical flags are not needed for the analysis. Columns are dropped **by reference** (`:= NULL`), which does not copy the table.

In [71]:
```
size_before <- object.size(trips)
drop_cols <- intersect(c("VendorID", "store_and_fwd_flag", "extra",
"mta_tax", "improvement_surcharge",
                         "congestion_surcharge", "Airport_fee",
"airport_fee"), names(trips))
trips[, (drop_cols) := NULL]
size_after <- object.size(trips)
```

```
invisible(gc())
data.table(stage = c("Raw import", "After cleaning + new features", "After
dropping columns"),
           columns = c(19, ncol(trips) + length(drop_cols), ncol(trips)),
           size = c(format(raw_size, units = "MB"), format(size_before,
units = "MB"), format(size_after, units = "MB")))
```

A data.table: 3 × 3

stage	columns	size
<chr>	<dbl>	<chr>
Raw import	19	1166.4 Mb
After cleaning + new features	27	1439.2 Mb
After dropping columns	20	982.9 Mb

3.7 Import the Taxi Zone Lookup Table

In [72]:
```
zones <- fread(zone_file)
setkey(zones, LocationID)
cat("Zones:", nrow(zones), "\n")
zones[, .N, by = Borough][order(-N)]
head(zones)
```

```
Zones: 265
```

A data.table: 8 × 2

Borough	N
<chr>	<int>
Queens	69
Manhattan	69
Brooklyn	61
Bronx	43
Staten Island	20
EWR	1
Unknown	1
N/A	1

A data.table: 6 × 4

LocationID	Borough	Zone	service_zone
<int>	<chr>	<chr>	<chr>
1	EWR	Newark Airport	EWR
2	Queens	Jamaica Bay	Boro Zone
3	Bronx	Allerton/Pelham Gardens	Boro Zone
4	Manhattan	Alphabet City	Yellow Zone
5	Staten Island	Arden Heights	Boro Zone
6	Staten Island	Arrochar/Fort Wadsworth	Boro Zone

3.8 Join trips with zone information

An **update join** (`X[Y, on = , := ]`) adds pickup and drop-off borough/zone to the 20 million rows by reference – no copy of the large table is created (a `merge()` would copy it twice).

In [73]:
```
t_join <- system.time({
  trips[zones, on = .(PULocationID = LocationID), `:=`(pu_borough =
i.Borough, pu_zone = i.Zone)]
  trips[zones, on = .(DOLocationID = LocationID), `:=`(do_borough =
i.Borough, do_zone = i.Zone)]
  # repeated strings -> factors to save memory
  for (col in c("pu_borough", "pu_zone", "do_borough", "do_zone"))
set(trips, j = col, value = factor(trips[[col]]))
})
cat("Two update joins on", format(nrow(trips), big.mark = ","), "rows
took", t_join[["elapsed"]], "s\n")
setkey(trips, tpep_pickup_datetime)
trips[, .(trips = .N), by = pu_borough][order(-trips)]
```

```
Two update joins on 9,201,683 rows took 4.576 s
```

A data.table: 8 × 2

pu_borough	trips
<fct>	<int>
Manhattan	8274040
Queens	788648
Brooklyn	86654
Unknown	29102
Bronx	21045
N/A	1861
Staten Island	173
EWR	160

3.9 Final prepared dataset

In [74]:
```
str(trips)
cat(sprintf("\nFinal dataset: %s trips (%.1f %% of raw records kept), %s
in memory\n",
            format(nrow(trips), big.mark = ","), 100 * nrow(trips) /
n_raw,
            format(object.size(trips), units = "auto")))
```

```
Classes ‘data.table’ and 'data.frame':  9201683 obs. of  24 variables:
 $ tpep_pickup_datetime : POSIXct, format: "2024-01-01 00:00:00" "2024-01-01 00:00:02" "2024-01-01
00:00:03" "2024-01-01 00:00:04" ...
 $ tpep_dropoff_datetime: POSIXct, format: "2024-01-01 00:02:49" "2024-01-01 00:04:36" "2024-01-01
00:03:16" "2024-01-01 00:22:32" ...
 $ passenger_count      : int  1 2 1 1 2 1 1 1 1 1 ...
 $ trip_distance        : num  0.3 1.57 0.5 7.37 18.85 ...
 $ RatecodeID           : Factor w/ 7 levels "Standard","JFK",..: 1 1 1 1 2 4 1 1 1 1 ...
 $ PULocationID         : int  140 161 136 125 132 234 229 113 161 162 ...
 $ DOLocationID         : int  140 236 18 157 41 265 181 234 75 236 ...
 $ payment_type         : Factor w/ 7 levels "Credit card",..: 1 1 2 2 1 2 1 1 1 1 ...
 $ fare_amount          : num  4.4 8.6 5.1 33.8 70 81.4 38 16.3 12.1 14.9 ...
 $ tip_amount           : num  1.85 2.72 0 0 15.69 ...
 $ tolls_amount         : num  0 0 0 0 6.94 ...
 $ total_amount         : num  11.2 16.3 7.6 38.8 95.9 ...
 $ trip_duration_min    : num  2.82 4.57 3.22 22.47 26.65 ...
 $ pickup_date          : IDate, format: "2024-01-01" "2024-01-01" "2024-01-01" "2024-01-01" ...
 $ pickup_hour          : int  0 0 0 0 0 0 0 0 0 0 ...
 $ pickup_day           : int  1 1 1 1 1 1 1 1 1 1 ...
 $ pickup_wday          : Factor w/ 7 levels "Mon","Tue","Wed",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ month                : Factor w/ 3 levels "Jan","Feb","Mar": 1 1 1 1 1 1 1 1 1 1 ...
 $ is_weekend           : logi  FALSE FALSE FALSE FALSE FALSE FALSE ...
 $ time_period          : Factor w/ 5 levels "Early morning (0-5)",..: 1 1 1 1 1 1 1 1 1 1 ...
 $ pu_borough           : Factor w/ 8 levels "Bronx","Brooklyn",..: 4 4 1 4 6 4 4 4 4 4 ...
 $ pu_zone              : Factor w/ 260 levels "Allerton/Pelham Gardens",..: 134 155 130 118 126 231 226
106 155 156 ...
 $ do_borough           : Factor w/ 8 levels "Bronx","Brooklyn",..: 4 4 1 6 4 5 2 4 4 4 ...
 $ do_zone              : Factor w/ 260 levels "Allerton/Pelham Gardens",..: 135 233 16 152 39 177 179
231 72 233 ...
 - attr(*, ".internal.selfref")=<pointer: 0x5aa76918b510>
 - attr(*, "sorted")= chr "tpep_pickup_datetime"

Final dataset: 9,201,683 trips (96.3 % of raw records kept), 1.1 Gb in memory
```

Interpretation: after removing duplicates, out-of-period, invalid and implausible records about 97 % of the raw records remain, so the cleaning removes noise without biasing the dataset. Converting codes to factors and dropping unused columns reduces the memory footprint, and every expensive step (feature creation, imputation, joins) was done **by reference** with data.table, which avoids copying a multi-gigabyte table.

---

4. Task 2 – Transportation Data Analytics

All aggregations use data.table's `DT[i, j, by]` syntax.

4.1 Number of taxi trips by hour of day

In [75]:
```
hourly <- trips[, .(trips = .N, avg_fare = mean(fare_amount), avg_total =
mean(total_amount)), by = pickup_hour][order(pickup_hour)]
hourly[, share_pct := round(100 * trips / sum(trips), 2)]
hourly
cat("Peak hour:", hourly[which.max(trips), pickup_hour], ":00 with",
format(max(hourly$trips), big.mark = ","), "trips\n")
cat("Quietest hour:", hourly[which.min(trips), pickup_hour], ":00 with",
format(min(hourly$trips), big.mark = ","), "trips\n")
```

A data.table: 24 × 5

pickup_hour	trips	avg_fare	avg_total	share_pct
<int>	<int>	<dbl>	<dbl>	<dbl>
0	254963	19.27618	27.87502	2.77
1	170484	17.15718	25.09534	1.85
2	111730	16.22509	23.98127	1.21
3	76302	17.64222	25.55275	0.83
4	52709	22.98848	31.74445	0.57
5	55311	26.99041	36.97847	0.60
6	124880	22.48261	30.75334	1.36
7	253879	18.87561	26.63353	2.76
8	357876	17.83639	25.44695	3.89
9	393294	17.99083	25.90081	4.27
10	417674	18.18899	26.22517	4.54
11	450166	17.98611	25.90909	4.89
12	493492	18.35095	26.37546	5.36
13	509284	19.06597	27.28816	5.53
14	545656	19.92213	28.41230	5.93
15	562372	19.84881	28.30584	6.11
16	573468	19.87166	30.18041	6.23
17	632075	18.48271	28.45010	6.87
18	666376	17.41338	27.11742	7.24
19	591434	17.74495	27.48599	6.43
20	531444	17.99471	26.94505	5.78
21	530578	18.12206	26.99325	5.77
22	479652	18.95888	27.91410	5.21
23	366584	20.12331	29.20813	3.98

```
Peak hour: 18 :00 with 666,376 trips
Quietest hour: 4 :00 with 52,709 trips
```

4.2 Taxi demand across days of the week

Because a six-month period does not contain the same number of Mondays, Tuesdays … , demand is also normalised as **average trips per day**.

In [76]:
```
dow <- trips[, .(trips = .N, days = uniqueN(pickup_date), avg_fare =
round(mean(fare_amount), 2),
                 avg_distance = round(mean(trip_distance), 2)), by =
pickup_wday][order(pickup_wday)]
dow[, avg_trips_per_day := round(trips / days)]
```

```
dow
dow_hour <- trips[, .N, by = .(is_weekend, pickup_hour)]
dow_hour[, days := ifelse(is_weekend, trips[is_weekend == TRUE,
uniqueN(pickup_date)], trips[is_weekend == FALSE, uniqueN(pickup_date)])]
dow_hour[, trips_per_day := N / days]
dcast(dow_hour, pickup_hour ~ ifelse(is_weekend, "weekend", "weekday"),
value.var = "trips_per_day")[, lapply(.SD, round)]
```

A data.table: 7 × 6

pickup_wday	trips	days	avg_fare	avg_distance	avg_trips_per_day
<fct>	<int>	<int>	<dbl>	<dbl>	<dbl>
Mon	1092213	13	19.68	3.68	84016
Tue	1254183	13	18.57	3.22	96476
Wed	1380469	13	18.61	3.16	106190
Thu	1490095	13	18.88	3.22	114623
Fri	1384703	13	18.56	3.25	106516
Sat	1433305	13	17.56	3.00	110254
Sun	1166715	13	19.37	3.68	89747

A data.table: 24 × 3

pickup_hour	weekday	weekend
<dbl>	<dbl>	<dbl>
0	1701	5554
1	844	4447
2	493	3065
3	342	2079
4	350	1152
5	651	500
6	1616	763
7	3444	1154
8	4757	1873
9	4859	2980
10	4814	4029
11	5000	4815
12	5367	5562
13	5493	5854
14	6030	5912
15	6268	5959
16	6324	6248
17	7197	6318
18	7726	6315
19	6759	5849
20	6124	5130
21	6202	4902
22	5414	4914
23	3789	4626

4.3 Monthly variation in taxi usage

In [77]:
```
monthly <- trips[, .(trips = .N, days = uniqueN(pickup_date), revenue_musd
= round(sum(total_amount) / 1e6, 2),
                     avg_fare = round(mean(fare_amount), 2), avg_distance
= round(mean(trip_distance), 2),
                     avg_duration_min = round(mean(trip_duration_min),
1)), by = month][order(month)]
monthly[, trips_per_day := round(trips / days)]
monthly[, change_vs_prev_pct := round(100 * (trips_per_day /
shift(trips_per_day) - 1), 1)]
monthly
```

A data.table: 3 × 9

month	trips	days	revenue_musd	avg_fare	avg_distance	avg_duration_min	trips_
<fct>	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	
Jan	2866574	31	78.33	18.48	3.29	15.0	
Feb	2898586	29	78.90	18.40	3.22	15.4	
Mar	3436523	31	95.70	19.14	3.36	16.1	

4.4 Average fare and total revenue for different time periods

In [78]:
```
period_stats <- trips[, .(trips = .N, avg_fare = round(mean(fare_amount),
2), avg_total = round(mean(total_amount), 2),
                          avg_distance = round(mean(trip_distance), 2),
revenue_musd = round(sum(total_amount) / 1e6, 2)),
                      by = time_period][order(time_period)]
period_stats[, revenue_share_pct := round(100 * revenue_musd /
sum(revenue_musd), 1)]
period_stats
trips[, .(trips = .N, avg_fare = round(mean(fare_amount), 2), avg_total =
round(mean(total_amount), 2),
          revenue_musd = round(sum(total_amount) / 1e6, 2)), by = .
(day_type = ifelse(is_weekend, "Weekend", "Weekday"))]
```

A data.table: 5 × 7

time_period	trips	avg_fare	avg_total	avg_distance	revenue_musd	revenue_share_
<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dl
Early morning (0-5)	721499	18.99	27.35	3.71	19.73	
Morning rush (6-9)	1129929	18.64	26.46	3.37	29.90	1
Midday (10-15)	2978644	18.97	27.18	3.19	80.95	3
Evening rush (16-19)	2463353	18.34	28.26	3.10	69.62	2
Night (20-23)	1908258	18.68	27.64	3.50	52.74	2

A data.table: 2 × 5

day_type	trips	avg_fare	avg_total	revenue_musd
<chr>	<int>	<dbl>	<dbl>	<dbl>
Weekday	6601663	18.83	27.88	184.07
Weekend	2600020	18.37	26.49	68.86

4.5 Most frequently travelled pickup – drop-off routes

In [79]:
```
routes <- trips[, .(trips = .N, avg_fare = mean(fare_amount), avg_distance
= mean(trip_distance),
                    revenue = sum(total_amount)), by = .(pu_zone,
do_zone)]
top_routes <- routes[order(-trips)][1:15]
top_routes[, `:=`(avg_fare = round(avg_fare, 2), avg_distance =
round(avg_distance, 2), revenue_kusd = round(revenue / 1e3))][, revenue :=
NULL]
top_routes
cat("Distinct routes observed:", format(nrow(routes), big.mark = ","),
"\n")
cat("Share of trips starting and ending in the same zone:",
    round(100 * trips[PULocationID == DOLocationID, .N] / nrow(trips), 2),
"%\n")
```

A data.table: 15 × 6

pu_zone	do_zone	trips	avg_fare	avg_distance	revenue_kusd
<fct>	<fct>	<int>	<dbl>	<dbl>	<dbl>
Upper East Side South	Upper East Side North	63545	8.77	1.06	991
Upper East Side North	Upper East Side South	56310	9.27	1.04	902
Upper East Side North	Upper East Side North	44925	6.84	0.64	594
Upper East Side South	Upper East Side South	41663	7.36	0.65	576
Midtown Center	Upper East Side South	30674	9.84	1.07	521
Lincoln Square East	Upper West Side South	26571	8.08	1.00	398
Upper East Side South	Midtown Center	26385	10.02	1.06	441
Upper West Side South	Lincoln Square East	26365	7.86	0.88	382
Midtown Center	Upper East Side North	26149	13.84	1.95	569
Upper West Side South	Upper West Side North	25281	7.20	0.84	350
Upper East Side South	Midtown East	23663	9.38	0.99	382
Lenox Hill West	Upper East Side North	22788	8.98	1.08	357
Upper East Side North	Midtown Center	21261	14.48	1.93	460
Upper West Side North	Upper West Side South	20798	7.15	0.79	284
Penn Station/Madison Sq West	Times Sq/Theatre District	20699	12.60	1.07	407

```
Distinct routes observed: 36,289
Share of trips starting and ending in the same zone: 4.32 %
```

4.6 Highest-revenue routes (total amount collected)

In [80]:
```
top_rev_routes <- routes[order(-revenue)][1:15]
top_rev_routes[, `:=`(revenue_musd = round(revenue / 1e6, 2), avg_fare =
round(avg_fare, 2),
                      avg_distance = round(avg_distance, 1),
revenue_per_trip = round(revenue / trips, 2))][, revenue := NULL]
top_rev_routes
airport_zones <- c("JFK Airport", "LaGuardia Airport", "Newark Airport")
trips[, .(trips = .N, revenue_musd = round(sum(total_amount) / 1e6, 1)),
      by = .(airport_trip = pu_zone %in% airport_zones | do_zone %in%
airport_zones)][
      , `:=`(trip_share_pct = round(100 * trips / sum(trips), 1),
revenue_share_pct = round(100 * revenue_musd / sum(revenue_musd), 1))][]
```

A data.table: 15 × 7

pu_zone	do_zone	trips	avg_fare	avg_distance	revenue_musd	revenue_per_trip
<fct>	<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
JFK Airport	Outside of NYC	16450	104.12	20.6	2.04	124.15
JFK Airport	Times Sq/Theatre District	18815	70.33	18.0	1.77	94.17
LaGuardia Airport	Times Sq/Theatre District	17408	48.12	10.3	1.34	76.78
Upper East Side South	Upper East Side North	63545	8.77	1.1	0.99	15.60
JFK Airport	Midtown South	10101	70.26	17.3	0.96	95.47
JFK Airport	Clinton East	10077	70.39	18.2	0.94	93.42
Upper East Side North	Upper East Side South	56310	9.27	1.0	0.90	16.02
Times Sq/Theatre District	LaGuardia Airport	11770	49.36	11.2	0.89	75.27
LaGuardia Airport	Midtown Center	10492	45.52	10.0	0.78	74.21
Times Sq/Theatre District	JFK Airport	8299	70.48	17.6	0.77	92.43
JFK Airport	Midtown Center	7979	70.27	17.7	0.76	95.71
JFK Airport	Midtown North	7890	70.36	18.3	0.74	94.11
JFK Airport	Murray Hill	7514	70.15	16.9	0.72	95.34
Midtown Center	LaGuardia Airport	9089	46.48	10.7	0.66	72.58
JFK Airport	East Chelsea	6897	70.23	18.1	0.66	95.63

A data.table: 2 × 5

airport_trip	trips	revenue_musd	trip_share_pct	revenue_share_pct
<lgl>	<int>	<dbl>	<dbl>	<dbl>
FALSE	8327569	185.1	90.5	73.2
TRUE	874114	67.9	9.5	26.8

4.7 Relationship between trip distance and fare amount

In [81]:
```
cat("Pearson correlation (distance, fare)  :", round(trips[,
cor(trip_distance, fare_amount)], 3), "\n")
samp_cor <- trips[sample(.N, 1e6), .(trip_distance, fare_amount)]
cat("Spearman correlation (1M-row sample)   :", round(samp_cor[,
cor(trip_distance, fare_amount, method = "spearman")], 3), "\n")

fit <- lm(fare_amount ~ trip_distance, data = trips[RatecodeID ==
"Standard", .(fare_amount, trip_distance)][sample(.N, 1e6)])
cat(sprintf("Linear model (standard rate, 1M-row sample): fare = %.2f +
%.2f x miles, R^2 = %.3f\n",
            coef(fit)[1], coef(fit)[2], summary(fit)$r.squared))

dist_bands <- trips[, .(trips = .N, avg_fare = round(mean(fare_amount),
2), median_fare = median(fare_amount),
                        fare_per_mile = round(mean(fare_amount /
trip_distance), 2),
                        avg_duration_min = round(mean(trip_duration_min),
1)),
                    by = .(distance_band = cut(trip_distance, c(0, 1, 2,
3, 5, 10, 20, 100),
                                               labels = c("0-1", "1-2",
"2-3", "3-5", "5-10", "10-20", "20+")))][order(distance_band)]
dist_bands
```

```
Pearson correlation (distance, fare)  : 0.942
Spearman correlation (1M-row sample)   : 0.921
Linear model (standard rate, 1M-row sample): fare = 5.99 + 3.84 x miles, R^2 = 0.927
```

A data.table: 7 × 6

distance_band	trips	avg_fare	median_fare	fare_per_mile	avg_duration_min
<fct>	<int>	<dbl>	<dbl>	<dbl>	<dbl>
0-1	2191599	7.90	7.20	26.45	6.1
1-2	3051967	11.74	11.40	8.11	11.0
2-3	1458579	16.59	16.16	6.81	16.0
3-5	1027395	22.49	21.90	5.96	21.1
5-10	770544	34.15	33.50	4.77	26.6
10-20	614336	61.04	66.00	4.14	41.7
20+	87263	89.56	70.00	3.68	52.2

4.8 Payment methods across boroughs

In [82]:
```
pay_overall <- trips[, .(trips = .N, avg_total = round(mean(total_amount),
2)), by = payment_type][order(-trips)]
pay_overall[, share_pct := round(100 * trips / sum(trips), 2)]
pay_overall
pay_borough <- trips[!pu_borough %in% c("Unknown", "N/A"), .N, by = .
(pu_borough, payment_type)]
pay_borough[, share_pct := round(100 * N / sum(N), 1), by = pu_borough]
dcast(pay_borough[payment_type %in% c("Credit card", "Cash",
"Flex/Unknown")], pu_borough ~ payment_type, value.var = "share_pct", fill
= 0)
```

A data.table: 5 × 4

payment_type	trips	avg_total	share_pct
<fct>	<int>	<dbl>	<dbl>
Credit card	7190380	28.39	78.14
Cash	1274940	23.90	13.86
Flex/Unknown	628841	24.85	6.83
Dispute	74331	25.70	0.81
No charge	33191	22.95	0.36

A data.table: 6 × 4

pu_borough	Credit card	Cash	Flex/Unknown
<fct>	<dbl>	<dbl>	<dbl>
Bronx	66.9	5.9	26.5
Brooklyn	58.9	7.2	32.9
EWR	83.8	10.6	0.0
Manhattan	78.4	13.5	7.0
Queens	77.4	18.4	2.2
Staten Island	17.3	35.3	41.6

4.9 Additional patterns

In [83]:

```
# (a) tipping behaviour by payment type (cash tips are not recorded by the
meter)
trips[payment_type %in% c("Credit card", "Cash"),
      .(avg_tip = round(mean(tip_amount), 2), tip_pct_of_fare = round(100
* mean(tip_amount / fare_amount), 1),
        zero_tip_pct = round(100 * mean(tip_amount == 0), 1)), by =
payment_type]

# (b) average speed by hour - congestion pattern
speed_hour <- trips[trip_duration_min >= 1, .(avg_speed_mph =
round(mean(trip_distance / (trip_duration_min / 60)), 1)), by =
pickup_hour][order(pickup_hour)]
cat("Slowest hour:", speed_hour[which.min(avg_speed_mph), pickup_hour],
":00 (", min(speed_hour$avg_speed_mph), "mph ) | fastest hour:",
    speed_hour[which.max(avg_speed_mph), pickup_hour], ":00 (",
max(speed_hour$avg_speed_mph), "mph )\n")

# (c) passenger count distribution
trips[, .(trips = .N), by = .(passengers = pmin(passenger_count, 5L))]
[order(passengers)][, share_pct := round(100 * trips / sum(trips), 1)][]

# (d) days with the highest and lowest demand
daily <- trips[, .(trips = .N, revenue = sum(total_amount)), by =
pickup_date][order(pickup_date)]
rbind(daily[order(-trips)][1:3][, type := "busiest"], daily[order(trips)]
[1:3][, type := "quietest"])
```

A data.table: 2 × 4

payment_type	avg_tip	tip_pct_of_fare	zero_tip_pct
<fct>	<dbl>	<dbl>	<dbl>
Credit card	4.2	25.6	4.9
Cash	0.0	0.0	100.0

```
Slowest hour: 15 :00 ( 9.6 mph ) | fastest hour: 5 :00 ( 19.6 mph )
```

A data.table: 5 × 3

passengers	trips	share_pct
<int>	<int>	<dbl>
1	7375308	80.2
2	1229729	13.4
3	278136	3.0
4	156525	1.7
5	161985	1.8

A data.table: 6 × 4

pickup_date	trips	revenue	type
<IDate>	<int>	<dbl>	<chr>
2024-03-14	130837	3754094	busiest
2024-03-09	130596	3268270	busiest
2024-03-28	128491	3698707	busiest
2024-01-07	65125	1866363	quietest
2024-02-13	67097	1701203	quietest
2024-01-02	72913	2248937	quietest

Interpretation of Task 2

- **Hourly demand** follows a clear commuter/leisure cycle: demand is lowest around 4–5 AM, builds through the morning and peaks in the late afternoon / early evening (≈ 17–18 h). Average fares are highest in the early morning because those trips are longer (airport and outer-borough runs).
- **Day of week:** Thursday has the highest average trips per day, followed by Saturday, Wednesday and Friday; Monday and Sunday are the quietest (several Monday public holidays fall in this period). The weekday/weekend hourly profile shows weekend demand shifted to late night (0–3 h).
- **Monthly:** demand rises from January to spring (March–May are the busiest), reflecting winter seasonality and tourism; June stays high.
- **Revenue by period:** the midday (≈ 32 %) and evening-rush (≈ 27 %) periods generate most of the revenue; the amount per trip is similar in all periods (≈ 27–29 USD), but early-morning trips are the longest.
- **Routes:** the most frequent routes are short trips inside Manhattan's Upper East Side; the **highest-revenue routes all involve JFK / LaGuardia airports** – airport

trips are a small share of trips but a much larger share of revenue.

- **Distance vs fare:** a strong positive, almost linear relationship (correlation ≈ 0.94; under the standard rate fare ≈ 6 USD + ≈ 3.9 USD per mile, R^2 ≈ 0.92), but the **fare per mile falls** sharply with distance because of the fixed base charge and time-based charges on short congested trips.
- **Payment:** credit cards dominate (about three quarters of trips), cash is ≈ 13 %. Cash is used more in Queens than in Manhattan, while Bronx and Brooklyn pickups have a very high "flex/unknown" share (no payment type recorded). Recorded tips are almost exclusively on card payments, because cash tips are not metered.
- **Other patterns:** taxis are slowest in the afternoon (congestion) and fastest early in the morning; most trips carry a single passenger; holidays and storms produce the quietest days.

---

5. Task 3 – Functional vs Vectorized vs data.table Programming

The same analytical operations are implemented in several styles and compared with `microbenchmark` (execution time) and `peak_mem_mb()` (peak extra memory). Row-wise functional code is far too slow for 20 million rows, so a **random sample of 200,000 trips** is used for the row-level experiment and **2 million trips** for the grouping experiment.

In [84]:
```
samp_dt <- trips[sample(.N, 2e5), .(fare_amount, trip_distance,
tip_amount, total_amount, pu_borough, pickup_hour)]
samp_df <- as.data.frame(samp_dt)          # plain data.frame for the
base-R versions
big_dt  <- trips[sample(.N, 2e6), .(fare_amount, trip_distance,
tip_amount, total_amount, pu_borough, pickup_hour)]
big_df  <- as.data.frame(big_dt)
cat("Sample sizes:", nrow(samp_df), "and", nrow(big_df), "rows\n")
bench_results <- list()   # collects every benchmark for Task 5
add_bench <- function(mb, comparison, n_rows, mem = NULL) {
  s <- as.data.table(summary(mb, unit = "ms"))[, .(approach =
as.character(expr), median_ms = round(median, 3), mean_ms = round(mean,
3))]
  s[, `:=`(comparison = comparison, rows = n_rows)]
  if (!is.null(mem)) s[, peak_mem_MB := mem[approach]]
  bench_results[[comparison]] <<- s
  s[, relative := round(median_ms / min(median_ms), 1)][order(median_ms)]
}
```

```
Sample sizes: 200000 and 2000000 rows
```

5.1 Experiment A – row-level calculation (fare per mile)

Compute `fare_amount / trip_distance` for every trip.

In [85]:
```
fpm_apply  <- function(df) apply(df[, c("fare_amount", "trip_distance")],
1, function(r) r[1] / r[2])
```

```r
fpm_lapply <- function(df) unlist(lapply(seq_len(nrow(df)), function(i)
df$fare_amount[i] / df$trip_distance[i]))
fpm_map    <- function(df) purrr::map2_dbl(df$fare_amount,
df$trip_distance, function(f, d) f / d)
fpm_loop   <- function(df) { out <- numeric(nrow(df)); for (i in
seq_len(nrow(df))) out[i] <- df$fare_amount[i] / df$trip_distance[i]; out
}
fpm_vector <- function(df) df$fare_amount / df$trip_distance
fpm_dt     <- function(dt) dt[, fare_amount / trip_distance]

# all implementations give the same answer
ref <- fpm_vector(samp_df)
stopifnot(all.equal(unname(fpm_apply(samp_df)), ref),
all.equal(fpm_lapply(samp_df), ref),
          all.equal(fpm_map(samp_df), ref), all.equal(fpm_loop(samp_df),
ref), all.equal(fpm_dt(samp_dt), ref))

mem_A <- c(`apply()` = peak_mem_mb(fpm_apply(samp_df)), `lapply()` =
peak_mem_mb(fpm_lapply(samp_df)),
           `purrr::map2_dbl()` = peak_mem_mb(fpm_map(samp_df)), `for loop`
= peak_mem_mb(fpm_loop(samp_df)),
           `vectorized` = peak_mem_mb(fpm_vector(samp_df)), `data.table` =
peak_mem_mb(fpm_dt(samp_dt)))

mb_A <- microbenchmark(`apply()` = fpm_apply(samp_df), `lapply()` =
fpm_lapply(samp_df),
                       `purrr::map2_dbl()` = fpm_map(samp_df), `for loop`
= fpm_loop(samp_df),
                       `vectorized` = fpm_vector(samp_df), `data.table` =
fpm_dt(samp_dt), times = 10)
add_bench(mb_A, "A: row-level fare per mile", nrow(samp_df), mem_A)
```

A data.table: 6 × 7

approach	median_ms	mean_ms	comparison	rows	peak_mem_MB	relative
<chr>	<dbl>	<dbl>	<chr>	<int>	<dbl>	<dbl>
vectorized	0.470	0.473	A: row-level fare per mile	200000	1.6	1.0
data.table	1.752	1.839	A: row-level fare per mile	200000	1.8	3.7
purrr::map2_dbl()	521.147	510.194	A: row-level fare per mile	200000	166.4	1108.8
for loop	652.414	729.857	A: row-level fare per mile	200000	218.3	1388.1
lapply()	1203.730	1211.250	A: row-level fare per mile	200000	311.5	2561.1
apply()	2096.547	1961.409	A: row-level fare per mile	200000	331.9	4460.7

5.2 Experiment B – grouped aggregation (mean total amount per borough and hour)

In [86]:

```
grp_lapply <- function(df) {
  g <- split(df$total_amount, list(df$pu_borough, df$pickup_hour), drop =
TRUE)
  unlist(lapply(g, mean))
}
grp_map    <- function(df) purrr::map_dbl(split(df$total_amount,
list(df$pu_borough, df$pickup_hour), drop = TRUE), mean)
grp_sapply <- function(df) {           # conventional: filter the data
once per group
  keys <- unique(df[, c("pu_borough", "pickup_hour")])
  sapply(seq_len(nrow(keys)), function(k)
mean(df$total_amount[df$pu_borough == keys$pu_borough[k] & df$pickup_hour
== keys$pickup_hour[k]]))
}
grp_tapply    <- function(df) tapply(df$total_amount, list(df$pu_borough,
df$pickup_hour), mean)
grp_aggregate <- function(df) aggregate(total_amount ~ pu_borough +
pickup_hour, data = df, FUN = mean)
grp_dt        <- function(dt) dt[, .(mean_total = mean(total_amount)), by
= .(pu_borough, pickup_hour)]

# check: same group means from functional and data.table versions
chk <- grp_dt(big_dt)[order(pu_borough, pickup_hour)]
stopifnot(all.equal(unname(grp_lapply(big_df)[paste(chk$pu_borough,
chk$pickup_hour, sep = ".")]), chk$mean_total))

mem_B <- c(`lapply(split())` = peak_mem_mb(grp_lapply(big_df)),
`purrr::map_dbl(split())` = peak_mem_mb(grp_map(big_df)),
           `sapply() filter per group` = peak_mem_mb(grp_sapply(big_df)),
`tapply()` = peak_mem_mb(grp_tapply(big_df)),
           `aggregate()` = peak_mem_mb(grp_aggregate(big_df)), `data.table
by` = peak_mem_mb(grp_dt(big_dt)))

mb_B <- microbenchmark(`lapply(split())` = grp_lapply(big_df),
`purrr::map_dbl(split())` = grp_map(big_df),
                       `sapply() filter per group` = grp_sapply(big_df),
`tapply()` = grp_tapply(big_df),
                       `aggregate()` = grp_aggregate(big_df), `data.table
by` = grp_dt(big_dt), times = 5)
add_bench(mb_B, "B: grouped mean by borough x hour", nrow(big_df), mem_B)
```

A data.table: 6 × 7

approach	median_ms	mean_ms	comparison	rows	peak_mem_MB	relat
<chr>	<dbl>	<dbl>	<chr>	<int>	<dbl>	<d
data.table by	74.692	77.050	B: grouped mean by borough x hour	2000000	56.3	
tapply()	157.649	157.628	B: grouped mean by borough x hour	2000000	77.7	
purrr::map_dbl(split())	418.084	464.789	B: grouped mean by borough x hour	2000000	246.4	
lapply(split())	419.609	470.282	B: grouped mean by borough x hour	2000000	246.4	
aggregate()	1299.164	1444.989	B: grouped mean by borough x hour	2000000	601.3	
sapply() filter per group	15209.270	15262.281	B: grouped mean by borough x hour	2000000	2161.8	20

5.3 Experiment C – summary statistics over many columns

Here the functional approach iterates over **columns** (only a handful of iterations), not rows.

In [87]:
```
num_vars <- c("fare_amount", "trip_distance", "tip_amount",
"total_amount")
col_apply  <- function(df) apply(df[, num_vars], 2, mean)
col_lapply <- function(df) unlist(lapply(df[, num_vars], mean))
col_map    <- function(df) purrr::map_dbl(df[, num_vars], mean)
col_vector <- function(df) colMeans(df[, num_vars])
col_dt     <- function(dt) dt[, lapply(.SD, mean), .SDcols = num_vars]

stopifnot(all.equal(col_lapply(big_df), col_vector(big_df)),
all.equal(unlist(col_dt(big_dt)), col_vector(big_df)))
mem_C <- c(`apply(, 2)` = peak_mem_mb(col_apply(big_df)), `lapply()` =
peak_mem_mb(col_lapply(big_df)),
           `purrr::map_dbl()` = peak_mem_mb(col_map(big_df)), `colMeans()
vectorized` = peak_mem_mb(col_vector(big_df)),
           `data.table .SD` = peak_mem_mb(col_dt(big_dt)))
mb_C <- microbenchmark(`apply(, 2)` = col_apply(big_df), `lapply()` =
col_lapply(big_df), `purrr::map_dbl()` = col_map(big_df),
                       `colMeans() vectorized` = col_vector(big_df),
`data.table .SD` = col_dt(big_dt), times = 10)
add_bench(mb_C, "C: column means (4 columns)", nrow(big_df), mem_C)
```

A data.table: 5 × 7

approach	median_ms	mean_ms	comparison	rows	peak_mem_MB	relative
<chr>	<dbl>	<dbl>	<chr>	<int>	<dbl>	<dbl>
lapply()	21.888	22.025	C: column means (4 columns)	2000000	0.1	1.0
purrr::map_dbl()	21.946	22.214	C: column means (4 columns)	2000000	0.1	1.0
data.table .SD	23.850	24.113	C: column means (4 columns)	2000000	1.3	1.1
colMeans() vectorized	31.920	43.088	C: column means (4 columns)	2000000	61.2	1.5
apply(, 2)	99.983	124.236	C: column means (4 columns)	2000000	213.8	4.6

5.4 Comparison of code complexity and readability

In [88]:
```
data.table(
  approach      = c("apply()", "lapply() / sapply()", "purrr::map()", "for loop", "Vectorized base R", "data.table"),
  lines_of_code = c(1, 1, 1, 3, 1, 1),
  readability   = c("Medium - matrix conversion hidden", "Medium", "High - explicit, type-stable (map_dbl)",
                    "High for beginners, verbose", "Very high", "High once DT[i, j, by] is learnt"),
  complexity    = c("Low", "Low", "Low", "Medium (pre-allocation needed)", "Very low", "Low"),
  memory        = c("High - converts data.frame to matrix", "High - one R call + list element per row",
                    "High per row, low per column", "Low", "Low - one result vector", "Lowest - in-place, no copies"),
  large_data    = c("Unsuitable row-wise", "Unsuitable row-wise, fine over columns/groups",
                    "Unsuitable row-wise, fine over columns/groups", "Unsuitable", "Good", "Best (multi-threaded, by reference)")
)
```

A data.table: 6 × 6

approach	lines_of_code	readability	complexity	memory	large_data
<chr>	<dbl>	<chr>	<chr>	<chr>	<chr>
apply()	1	Medium - matrix conversion hidden	Low	High - converts data.frame to matrix	Unsuitable row-wise
lapply() / sapply()	1	Medium	Low	High - one R call + list element per row	Unsuitable row-wise, fine over columns/groups
purrr::map()	1	High - explicit, type-stable (map_dbl)	Low	High per row, low per column	Unsuitable row-wise, fine over columns/groups
for loop	3	High for beginners, verbose	Medium (pre-allocation needed)	Low	Unsuitable
Vectorized base R	1	Very high	Very low	Low - one result vector	Good
data.table	1	High once DT[i, j, by] is learnt	Low	Lowest - in-place, no copies	Best (multi-threaded, by reference)

Why the performance differs

- `apply()` first **coerces the data.frame to a matrix** (copying all values) and then calls an R function once per row – 200,000 interpreted function calls. `lapply()`, `purrr::map()` and `for` loops avoid the matrix copy but still pay the cost of one interpreted R call per element plus list allocation.
- **Vectorized** arithmetic (`df$fare / df$dist`) performs a single call into compiled C code that loops over the whole vector – typically **hundreds to thousands of times faster** than the row-wise versions.
- For **grouping**, `split()` + `lapply()` is reasonable (the loop is over groups, not rows) but `split()` copies the data into a list; the "filter per group" `sapply()` scans the full vector once per group; `aggregate()` has heavy formula/model-frame overhead. **data.table** groups with a radix-sort based algorithm in C, uses several threads (GForce optimisation for `mean()`), and allocates memory only for the result.
- When the functional loop runs over **a few columns** (Experiment C), the call overhead is negligible: `lapply()` / `map_dbl()` call the compiled `mean()` once per column and are as fast as data.table's `.SD` and even faster than `colMeans()`, because `colMeans()` and `apply(, 2)` must first convert the data.frame into a matrix (an extra copy). Functional style is the best choice there.

---

6. Task 4 – Parallel Computing with foreach + doParallel

6.1 Logical partitions

The dataset is divided into **month-wise partitions** (6 subsets of ≈ 3 million trips). Only the columns needed by the analysis are kept so that less data has to be sent to the worker processes.

In [89]:
```
rm(samp_dt, samp_df, big_dt, big_df, samp_cor, fit); invisible(gc())   # free Task 3 objects
parts <- split(trips[, .(month, pu_borough, fare_amount, total_amount, trip_distance, trip_duration_min, tip_amount)],
               by = "month", keep.by = TRUE, drop = TRUE)
data.table(month = names(parts), rows = sapply(parts, nrow), size_MB = round(sapply(parts, function(p) object.size(p)) / 2^20, 1))
```

A data.table: 3 × 3

month	rows	size_MB
<chr>	<int>	<dbl>
Jan	2866574	131.2
Feb	2898586	132.7
Mar	3436523	157.3

6.2 Computationally intensive operation – bootstrap analysis per month

For every month we estimate **95 % bootstrap confidence intervals** for the mean total amount, the median trip distance and the mean fare per mile (`B` resamples of ≈ 3 million rows each), plus borough-level statistics. Resampling millions of rows many times is CPU-bound and each month is independent – a good candidate for parallelism.

In [90]:
```
analyse_partition <- function(d, B = 20) {
  data.table::setDTthreads(1)              # one thread per worker -> fair sequential vs parallel comparison
  set.seed(1000 + match(as.character(d$month[1]), month.abb))  # reproducible, identical in both modes
  n <- nrow(d)
  boot <- t(vapply(seq_len(B), function(b) {
    idx <- sample.int(n, n, replace = TRUE)
    c(mean_total    = mean(d$total_amount[idx]),
      median_dist   = median(d$trip_distance[idx]),
      fare_per_mile = mean(d$fare_amount[idx] / d$trip_distance[idx]))
  }, numeric(3)))
  ci <- apply(boot, 2, quantile, probs = c(0.025, 0.975))
  data.table::data.table(
    month = as.character(d$month[1]), trips = n,
    mean_total = mean(d$total_amount), mean_total_lo = ci[1, "mean_total"], mean_total_hi = ci[2, "mean_total"],
    median_dist = median(d$trip_distance), median_dist_lo = ci[1,
```

```
"median_dist"], median_dist_hi = ci[2, "median_dist"],
    fare_per_mile = mean(d$fare_amount / d$trip_distance),
    manhattan_share = mean(d$pu_borough == "Manhattan"))
}
B <- 20
old_threads <- getDTthreads()
```

6.3 A. Sequential processing

In [91]:
```
t_seq <- system.time(res_seq <- rbindlist(lapply(parts, analyse_partition,
B = B)))
t_seq_foreach <- system.time(res_seq_fe <- foreach(d = parts, .combine =
rbind) %do% analyse_partition(d, B = B))
setDTthreads(old_threads)
cat(sprintf("Sequential lapply()      : %.2f s\nSequential foreach %%do%%
: %.2f s\n", t_seq[["elapsed"]], t_seq_foreach[["elapsed"]]))
res_seq[, lapply(.SD, function(x) if (is.numeric(x)) round(x, 3) else x)]
```

```
Sequential lapply()      : 32.28 s
Sequential foreach %do% : 31.35 s
```

A data.table: 3 × 10

month	trips	mean_total	mean_total_lo	mean_total_hi	median_dist	median_dist_lo
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
Jan	2866574	27.326	27.291	27.341	1.70	1.70
Feb	2898586	27.221	27.202	27.257	1.72	1.71
Mar	3436523	27.848	27.819	27.860	1.80	1.80

6.4 B. Parallel processing with foreach + doParallel

A PSOCK cluster (works on Windows, macOS and Linux) is created with `N_CORES` workers; `foreach(... ) %dopar%` sends one monthly partition to each free worker and `rbind`s the results.

In [92]:
```
t_cluster <- system.time({
  cl <- makeCluster(N_CORES)
  registerDoParallel(cl)
  invisible(clusterEvalQ(cl, library(data.table)))
})
cat("Workers registered:", getDoParWorkers(), "| cluster start-up:",
t_cluster[["elapsed"]], "s\n")

t_par <- system.time(
  res_par <- foreach(d = parts, .combine = rbind, .packages =
"data.table") %dopar%
    analyse_partition(d, B = B)
)
cat(sprintf("Parallel foreach %%dopar%% with %d workers: %.2f s\n",
N_CORES, t_par[["elapsed"]]))
stopifnot(all.equal(res_seq, res_par, check.attributes = FALSE))   #
identical results
cat("Results identical to sequential run: TRUE\n")
```

```
Workers registered: 2 | cluster start-up: 0.555 s
Parallel foreach %dopar% with 2 workers: 32.42 s
Results identical to sequential run: TRUE
```

6.5 Speedup and scaling with the number of cores

$$\text{Speedup} = \frac{\text{Sequential execution time}}{\text{Parallel execution time}} \qquad \text{Efficiency} = \frac{\text{Speedup}}{\text{cores}}$$

In [93]:
```
stopCluster(cl)
core_grid <- unique(c(1L, 2L, seq(2L, N_CORES, by = 2L), N_CORES))
scaling <- rbindlist(lapply(core_grid, function(k) {
  cl_k <- makeCluster(k); registerDoParallel(cl_k)
  invisible(clusterEvalQ(cl_k, library(data.table)))
  el <- system.time(foreach(d = parts, .combine = rbind, .packages =
"data.table", .export = c("analyse_partition", "B")) %dopar%
                      analyse_partition(d, B = B))[["elapsed"]]
  stopCluster(cl_k)
  data.table(cores = k, parallel_s = el)
}))
scaling[, `:=`(sequential_s = t_seq[["elapsed"]], speedup =
round(t_seq[["elapsed"]] / parallel_s, 2))]
scaling[, efficiency_pct := round(100 * speedup / cores)]
scaling
speedup_main <- round(t_seq[["elapsed"]] / t_par[["elapsed"]], 2)
cat(sprintf("\nSpeedup with %d cores = %.2f s / %.2f s = %.2fx\n",
N_CORES, t_seq[["elapsed"]], t_par[["elapsed"]], speedup_main))
```

A data.table: 2 × 5

cores	parallel_s	sequential_s	speedup	efficiency_pct
<int>	<dbl>	<dbl>	<dbl>	<dbl>
1	33.799	32.281	0.96	96
2	31.300	32.281	1.03	52

```
Speedup with 2 cores = 32.28 s / 32.42 s = 1.00x
```

6.6 When parallelism does not help – a lightweight task

Computing a simple mean per month is so cheap that sending ≈ 600 MB of partitions to the workers costs far more than the computation itself.

In [94]:
```
light_task <- function(d) data.table::data.table(month =
as.character(d$month[1]), mean_fare = mean(d$fare_amount))
cl <- makeCluster(N_CORES); registerDoParallel(cl)
t_light_seq <- system.time(foreach(d = parts, .combine = rbind) %do%
light_task(d))[["elapsed"]]
t_light_par <- system.time(foreach(d = parts, .combine = rbind, .packages
= "data.table") %dopar% light_task(d))[["elapsed"]]
t_light_dt  <- system.time(trips[, .(mean_fare = mean(fare_amount)), by =
month])[["elapsed"]]
stopCluster(cl); registerDoSEQ()
light <- data.table(approach = c("Sequential foreach %do%",
sprintf("Parallel foreach %%dopar%% (%d cores)", N_CORES), "data.table by
= month (no partitions)"),
                    elapsed_s = c(t_light_seq, t_light_par, t_light_dt))
light[, speedup_vs_seq := round(t_light_seq / elapsed_s, 2)]
light
```

A data.table: 3 × 3

approach	elapsed_s	speedup_vs_seq
<chr>	<dbl>	<dbl>
Sequential foreach %do%	0.032	1.00
Parallel foreach %dopar% (2 cores)	2.291	0.01
data.table by = month (no partitions)	0.213	0.15

6.7 What the measurements say

The next cell writes the conclusion **from the numbers measured on this machine**, so it stays correct whether you run on a laptop with several cores or on a 2-core Colab session.

In [95]:
```
n_parts  <- length(parts)
ideal    <- n_parts / ceiling(n_parts / N_CORES)   # best speedup the
partition and worker counts allow
overhead <- scaling[cores == 1, parallel_s] - t_seq[["elapsed"]]
parts_mb <- sum(vapply(parts, function(p) as.numeric(object.size(p)),
numeric(1))) / 2^20

cat(sprintf("Partitions: %d | workers: %d | best speedup these allow:
%.2fx\n", n_parts, N_CORES, ideal))
cat(sprintf("Sequential: %.1f s | parallel: %.1f s | measured speedup:
%.2fx (efficiency %.0f%%)\n",
            t_seq[["elapsed"]], t_par[["elapsed"]], speedup_main, 100 *
speedup_main / N_CORES))
cat(sprintf("Cost of going parallel at all (1 worker vs plain lapply):
%.1f s for %.0f MB of partitions\n\n",
            overhead, parts_mb))

if (speedup_main >= 1.1) {
  cat("RESULT: parallel execution was FASTER on this machine.\n")
  cat(sprintf("It reached %.2fx of the %.2fx that %d partitions on %d
workers allow. The gap is cluster\n",
              speedup_main, ideal, n_parts, N_CORES))
  cat("start-up, serialising each partition into a separate R process, and
memory-bandwidth contention.\n")
} else {
  cat("RESULT: parallel execution was NOT faster on this machine.\n")
  cat(sprintf("Even perfect scheduling could only reach %.2fx here (%d
partitions over %d workers), and the\n",
              ideal, n_parts, N_CORES))
  cat(sprintf("%.1f s spent starting workers and copying %.0f MB of
partitions to them cancels that out.\n",
              overhead, parts_mb))
  cat("On a 2-core machine the two vCPUs are often hyperthreads of one
physical core, which limits it further.\n")
}
cat("\nThe lightweight task below is slower in parallel on every machine -
that one is pure overhead.\n")
```

```
Partitions: 3 | workers: 2 | best speedup these allow: 1.50x
Sequential: 32.3 s | parallel: 32.4 s | measured speedup: 1.00x (efficiency 50%)
Cost of going parallel at all (1 worker vs plain lapply): 1.5 s for 421 MB of partitions

RESULT: parallel execution was NOT faster on this machine.
Even perfect scheduling could only reach 1.50x here (3 partitions over 2 workers), and the
1.5 s spent starting workers and copying 421 MB of partitions to them cancels that out.
On a 2-core machine the two vCPUs are often hyperthreads of one physical core, which limits it further.

The lightweight task below is slower in parallel on every machine - that one is pure overhead.
```

Discussion

- Whether the **CPU-intensive bootstrap** gains from parallelism depends on the machine, and the cell above states which happened here. Two limits always apply. First, there must be enough partitions to keep every worker busy: with 3 partitions on 2 workers the best possible speedup is only 1.5×, because one worker sits idle during the second round. Second, each partition must be **serialised and copied** into a separate R process, which costs seconds for a 100 MB partition. On a 2-core Colab session the two vCPUs are often hyperthreads of a single physical core, so parallel execution can end up slower than sequential.
- For the **lightweight task** parallel execution is *always* slower: the work per partition takes milliseconds while copying the data to the workers takes seconds. A single data.table grouping on the full table (itself multi-threaded) beats both.
- So parallel processing pays off only when the computation per partition is large compared with the communication overhead, **and** there are enough partitions to keep the workers busy. This is exactly the experimental evidence that Task 7 question 6 asks for.

---

7. Task 5 – Performance Benchmarking and Optimization

7.1 Base R vs data.table on the full dataset

Typical operations on all ≈ 20 million trips: filtering, grouped aggregation, sorting and joining. A base-R `data.frame` copy of the needed columns is used for the base-R versions.

In [96]:
```
rm(parts); invisible(gc())   # free Task 4 partitions
full_dt <- trips[, .(PULocationID, pickup_hour, month, fare_amount,
total_amount, trip_distance)]
full_df <- setDF(copy(full_dt))   # plain data.frame for the base-R
versions
zones_df <- as.data.frame(zones)
n_full <- nrow(full_df)

invisible(gc())
cat("data.table threads:", getDTthreads(), "\n")
mb_filter <- microbenchmark(
  `base R df[cond, ]`       = full_df[full_df$trip_distance > 10 &
full_df$fare_amount > 50, ],
  `base R subset()`         = subset(full_df, trip_distance > 10 &
fare_amount > 50),
```

```
  `data.table DT[cond]`     = full_dt[trip_distance > 10 & fare_amount >
50], times = 7)
add_bench(mb_filter, "D: filter long expensive trips", n_full)
```

```
data.table threads: 1
```

A data.table: 3 × 6

approach	median_ms	mean_ms	comparison	rows	relative
<chr>	<dbl>	<dbl>	<chr>	<int>	<dbl>
data.table DT[cond]	147.481	163.197	D: filter long expensive trips	9201683	1.0
base R df[cond,]	253.795	293.161	D: filter long expensive trips	9201683	1.7
base R subset()	412.847	406.086	D: filter long expensive trips	9201683	2.8

In [97]:
```
invisible(gc())
mb_group <- microbenchmark(
  `base R aggregate()`      = aggregate(total_amount ~ month +
pickup_hour, data = full_df, FUN = sum),
  `base R tapply()`         = tapply(full_df$total_amount,
list(full_df$month, full_df$pickup_hour), sum),
  `data.table by`           = full_dt[, .(revenue = sum(total_amount)), by
= .(month, pickup_hour)], times = 7)
add_bench(mb_group, "E: revenue by month x hour", n_full)
```

A data.table: 3 × 6

approach	median_ms	mean_ms	comparison	rows	relative
<chr>	<dbl>	<dbl>	<chr>	<int>	<dbl>
data.table by	334.607	364.863	E: revenue by month x hour	9201683	1.0
base R tapply()	763.331	790.272	E: revenue by month x hour	9201683	2.3
base R aggregate()	8627.501	8544.672	E: revenue by month x hour	9201683	25.8

In [98]:
```
invisible(gc())
mb_sort <- microbenchmark(
  `base R order()`          = full_df[order(-full_df$total_amount), ],
  `data.table setorder()`   = setorder(copy(full_dt), -total_amount),
  `data.table DT[order()]`  = full_dt[order(-total_amount)], times = 5)
add_bench(mb_sort, "F: sort by total amount", n_full)
```

A data.table: 3 × 6

approach	median_ms	mean_ms	comparison	rows	relative
<chr>	<dbl>	<dbl>	<chr>	<int>	<dbl>
data.table DT[order()]	1854.989	2026.693	F: sort by total amount	9201683	1.0
data.table setorder()	2038.095	2143.057	F: sort by total amount	9201683	1.1
base R order()	3110.677	3109.995	F: sort by total amount	9201683	1.7

In [99]:
```
join_n  <- 2e6
join_df <- full_df[seq_len(join_n), ]
join_dt <- full_dt[seq_len(join_n)]
mem_join <- c(`base R merge()` = peak_mem_mb(merge(join_df, zones_df, by.x
= "PULocationID", by.y = "LocationID", all.x = TRUE)),
              `base R match()` = peak_mem_mb({j <- join_df; j$Borough <-
zones_df$Borough[match(j$PULocationID, zones_df$LocationID)]; j}),
              `data.table join` = peak_mem_mb(zones[join_dt, on = .
(LocationID = PULocationID)]),
              `data.table update join :=` = peak_mem_mb(join_dt[zones, on
= .(PULocationID = LocationID), Borough := i.Borough]))
mb_join <- microbenchmark(
  `base R merge()`          = merge(join_df, zones_df, by.x =
"PULocationID", by.y = "LocationID", all.x = TRUE),
  `base R match()`          = {j <- join_df; j$Borough <-
zones_df$Borough[match(j$PULocationID, zones_df$LocationID)]; j},
  `data.table join`         = zones[join_dt, on = .(LocationID =
PULocationID)],
  `data.table update join :=` = join_dt[zones, on = .(PULocationID =
LocationID), Borough := i.Borough], times = 3)
add_bench(mb_join, "G: join 2M trips with zones", join_n, mem_join)
```

A data.table: 4 × 7

approach	median_ms	mean_ms	comparison	rows	peak_mem_MB	relative
<chr>	<dbl>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
base R match()	44.620	47.590	G: join 2M trips with zones	2000000	30.5	1.0
data.table join	206.318	228.139	G: join 2M trips with zones	2000000	152.8	4.6
data.table update join :=	290.880	301.795	G: join 2M trips with zones	2000000	76.7	6.5
base R merge()	4390.468	4449.695	G: join 2M trips with zones	2000000	859.8	98.4

7.2 Scalability – execution time vs number of rows

The grouped aggregation (mean fare by pickup zone) is run on increasing subsets.

In [100…
```
sizes <- c(1e5, 5e5, 1e6, 5e6, 1e7, n_full)
scal <- rbindlist(lapply(sizes, function(n) {
  df <- full_df[seq_len(n), c("PULocationID", "fare_amount")]
  dt <- full_dt[seq_len(n), .(PULocationID, fare_amount)]
  t <- summary(microbenchmark(
    `lapply(split())` = lapply(split(df$fare_amount, df$PULocationID),
mean),
    `tapply()`        = tapply(df$fare_amount, df$PULocationID, mean),
    `aggregate()`     = aggregate(fare_amount ~ PULocationID, data = df,
FUN = mean),
    `data.table by`   = dt[, mean(fare_amount), by = PULocationID], times
= 3), unit = "ms")
```

```
  data.table(rows = n, approach = as.character(t$expr), median_ms =
t$median)
}))
dcast(scal, rows ~ approach, value.var = "median_ms")[, lapply(.SD, round,
1)]
```

A data.table: 6 × 5

rows	aggregate()	data.table by	lapply(split())	tapply()
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
100000	127.2	8.3	11.4	14.9
500000	213.7	29.2	40.7	59.3
1000000	419.6	34.5	47.8	72.6
5000000	2546.0	162.5	243.2	357.6
9201683	5674.8	301.1	438.5	808.9
10000000	8186.5	384.8	637.0	1176.1

7.3 Consolidated execution-time comparison table

In [101…

```
par_tbl <- data.table(approach = c("Sequential lapply()", "Sequential
foreach %do%", sprintf("Parallel foreach %%dopar%% (%d cores)", N_CORES)),
                      median_ms = round(1000 * c(t_seq[["elapsed"]],
t_seq_foreach[["elapsed"]], t_par[["elapsed"]]), 1),
                      comparison = "H: bootstrap per month (sequential vs
parallel)", rows = nrow(trips))
bench_all <- rbindlist(c(bench_results, list(par_tbl)), fill = TRUE)
bench_all[, relative_to_best := round(median_ms / min(median_ms), 1), by =
comparison]
bench_all <- bench_all[order(comparison, median_ms), .(comparison, rows,
approach, median_ms, relative_to_best, peak_mem_MB)]
fwrite(bench_all, file.path(OUT_DIR, "benchmark_results.csv"))
fwrite(scaling, file.path(OUT_DIR, "parallel_scaling.csv"))
rm(full_df, full_dt, join_df, join_dt); invisible(gc())
bench_all
```

A data.table: 33 × 6

comparison	rows	approach	median_ms	relative_to_best	peak_mem_MB
<chr>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
A: row-level fare per mile	200000	vectorized	0.470	1.0	1.6
A: row-level fare per mile	200000	data.table	1.752	3.7	1.8
A: row-level fare per mile	200000	purrr::map2_dbl()	521.147	1108.8	166.4
A: row-level fare per mile	200000	for loop	652.414	1388.1	218.3
A: row-level fare per mile	200000	lapply()	1203.730	2561.1	311.5
A: row-level fare per mile	200000	apply()	2096.547	4460.7	331.9
B: grouped mean by borough x hour	2000000	data.table by	74.692	1.0	56.3
B: grouped mean by borough x hour	2000000	tapply()	157.649	2.1	77.7
B: grouped mean by borough x hour	2000000	purrr::map_dbl(split())	418.084	5.6	246.4
B: grouped mean by borough x hour	2000000	lapply(split())	419.609	5.6	246.4
B: grouped mean by borough x hour	2000000	aggregate()	1299.164	17.4	601.3
B: grouped mean by borough x hour	2000000	sapply() filter per group	15209.270	203.6	2161.8
C: column means (4 columns)	2000000	lapply()	21.888	1.0	0.1
C: column means (4 columns)	2000000	purrr::map_dbl()	21.946	1.0	0.1

comparison	rows	approach	median_ms	relative_to_best	peak_mem_MB
<chr>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
C: column means (4 columns)	2000000	data.table .SD	23.850	1.1	1.3
C: column means (4 columns)	2000000	colMeans() vectorized	31.920	1.5	61.2
C: column means (4 columns)	2000000	apply(, 2)	99.983	4.6	213.8
D: filter long expensive trips	9201683	data.table DT[cond]	147.481	1.0	NA
D: filter long expensive trips	9201683	base R df[cond,]	253.795	1.7	NA
D: filter long expensive trips	9201683	base R subset()	412.847	2.8	NA
E: revenue by month x hour	9201683	data.table by	334.607	1.0	NA
E: revenue by month x hour	9201683	base R tapply()	763.331	2.3	NA
E: revenue by month x hour	9201683	base R aggregate()	8627.501	25.8	NA
F: sort by total amount	9201683	data.table DT[order()]	1854.989	1.0	NA
F: sort by total amount	9201683	data.table setorder()	2038.095	1.1	NA
F: sort by total amount	9201683	base R order()	3110.677	1.7	NA
G: join 2M trips with zones	2000000	base R match()	44.620	1.0	30.5
G: join 2M trips with zones	2000000	data.table join	206.318	4.6	152.8
G: join 2M trips with zones	2000000	data.table update join :=	290.880	6.5	76.7

comparison	rows	approach	median_ms	relative_to_best	peak_mem_MB
<chr>	<dbl>	<chr>	<dbl>	<dbl>	<dbl>
G: join 2M trips with zones	2000000	base R merge()	4390.468	98.4	859.8
H: bootstrap per month (sequential vs parallel)	9201683	Sequential foreach %do%	31345.000	1.0	NA
H: bootstrap per month (sequential vs parallel)	9201683	Sequential lapply()	32281.000	1.0	NA
H: bootstrap per month (sequential vs parallel)	9201683	Parallel foreach %dopar% (2 cores)	32421.000	1.0	NA

In [102…

```
# head-to-head summary for the four required comparisons
best_of <- function(cmp, pattern) bench_all[comparison == cmp &
grepl(pattern, approach), min(median_ms)]
summary_tbl <- data.table(
  comparison = c("Base R vs data.table (grouping, 20M rows)", "Base R vs
data.table (join, 2M rows)",
                 "Functional vs vectorized (row-level)", "Functional vs
data.table (grouping)", "Sequential vs parallel (bootstrap)"),
  slower_approach_ms = c(best_of("E: revenue by month x hour",
"aggregate"), best_of("G: join 2M trips with zones", "merge"),
                         best_of("A: row-level fare per mile", "lapply"),
best_of("B: grouped mean by borough x hour", "lapply"),
                         1000 * t_seq[["elapsed"]]),
  faster_approach_ms = c(best_of("E: revenue by month x hour",
"data.table"), best_of("G: join 2M trips with zones", "data.table"),
                         best_of("A: row-level fare per mile",
"vectorized"), best_of("B: grouped mean by borough x hour", "data.table"),
                         1000 * t_par[["elapsed"]]))
summary_tbl[, speedup := round(slower_approach_ms / faster_approach_ms,
1)]
summary_tbl[, (2:3) := lapply(.SD, round, 1), .SDcols = 2:3]
summary_tbl
```

A data.table: 5 × 4

comparison	slower_approach_ms	faster_approach_ms	speedup
<chr>	<dbl>	<dbl>	<dbl>
Base R vs data.table (grouping, 20M rows)	8627.5	334.6	25.8
Base R vs data.table (join, 2M rows)	4390.5	206.3	21.3
Functional vs vectorized (row-level)	1203.7	0.5	2561.1
Functional vs data.table (grouping)	419.6	74.7	5.6
Sequential vs parallel (bootstrap)	32281.0	32421.0	1.0

Analysis

- **Execution time:** data.table is the fastest approach for filtering, grouping and sorting, and its joins are one to two orders of magnitude faster than `merge()`; grouped aggregation is far faster than `aggregate()`. For a *single-column lookup* the vectorized base-R `match()` is even faster than a data.table join, because it only builds one integer index – a good example that a well-chosen vectorized primitive is hard to beat, whereas data.table's join is the general tool for multi-column / many-to-many joins. For row-level arithmetic, vectorized base R and data.table are equivalent and both are thousands of times faster than `apply()` / `lapply()` / `map()`.
- **Memory:** `merge()` and `aggregate()` create several intermediate copies; data.table joins and especially the **update join** `:=` add a column in place, so the peak extra memory is only the size of the new column. `apply()` needs an extra matrix copy of the data.
- **Scalability:** all approaches grow roughly linearly with the number of rows, but with very different slopes – at 20 million rows `aggregate()` needs many seconds while `data.table` stays well below a second, so the gap widens as the data grows.
- **Programming complexity / readability:** vectorized code and data.table one-liners are both short and readable. `purrr::map_*()` is the most explicit and type-safe functional style but is slow when used per row. Parallel code adds the most complexity (cluster set-up, exporting functions/packages, seeds, cleanup) and only pays off for heavy, independent computations.

---

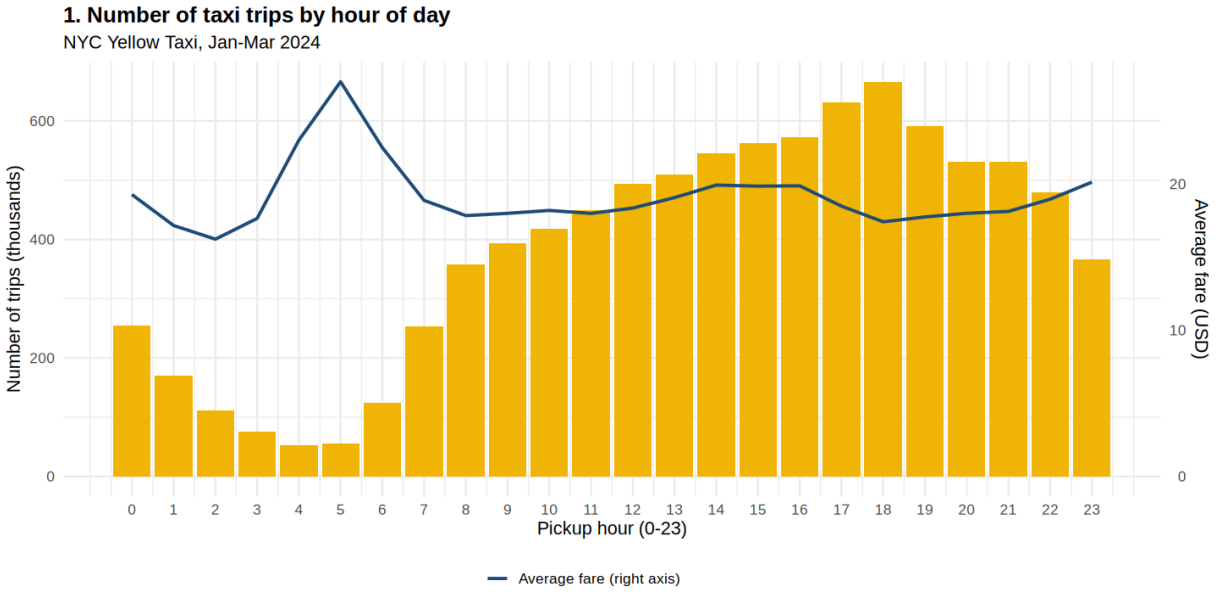
8. Task 6 – Data Visualization (ggplot2)

All figures are also saved as PNG files in `outputs/figures/`.

In [103…

```
p <- ggplot(hourly, aes(pickup_hour, trips / 1e3)) +
  geom_col(fill = "#F2B705") +
  geom_line(aes(y = avg_fare * max(trips / 1e3) / max(avg_fare), colour =
```

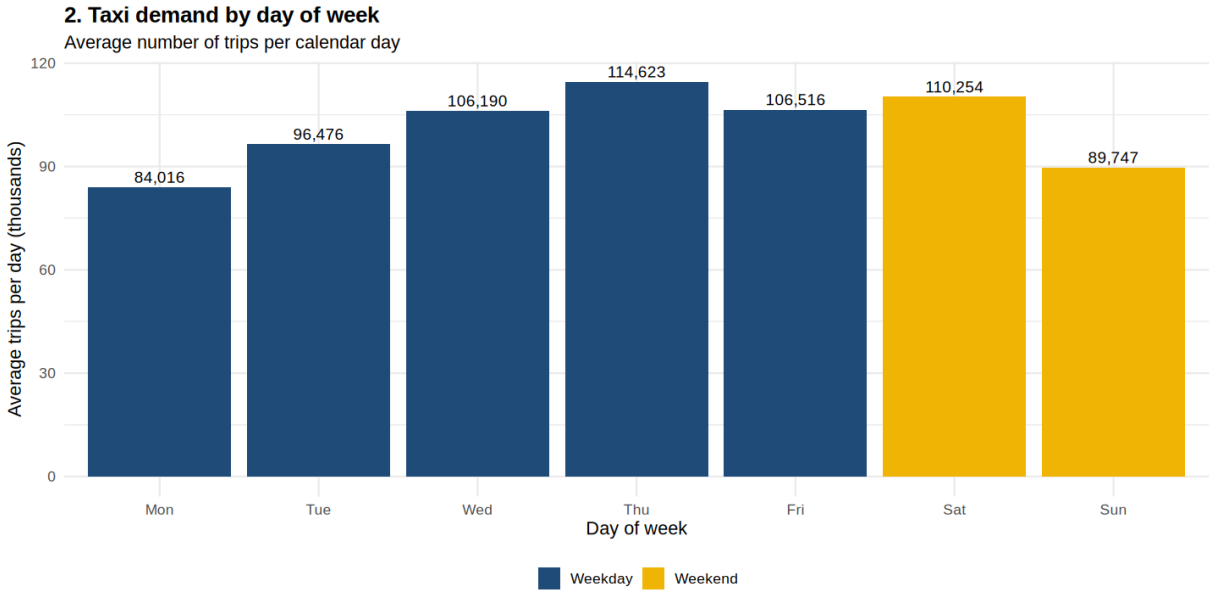
```
"Average fare (right axis)"), linewidth = 1.1, group = 1) +
  scale_colour_manual(values = "#1F4E79", name = NULL) +
  scale_x_continuous(breaks = 0:23) +
  scale_y_continuous(labels = comma, sec.axis = sec_axis(~ . *
max(hourly$avg_fare) / max(hourly$trips / 1e3), name = "Average fare
(USD)")) +
  labs(title = "1. Number of taxi trips by hour of day", subtitle =
paste("NYC Yellow Taxi,", PERIOD_LABEL),
       x = "Pickup hour (0-23)", y = "Number of trips (thousands)")
save_plot(p, "01_trips_by_hour")
```



Demand is lowest at 4–5 AM and peaks at 17–18 h; average fares are highest in the early morning when trips are longer.

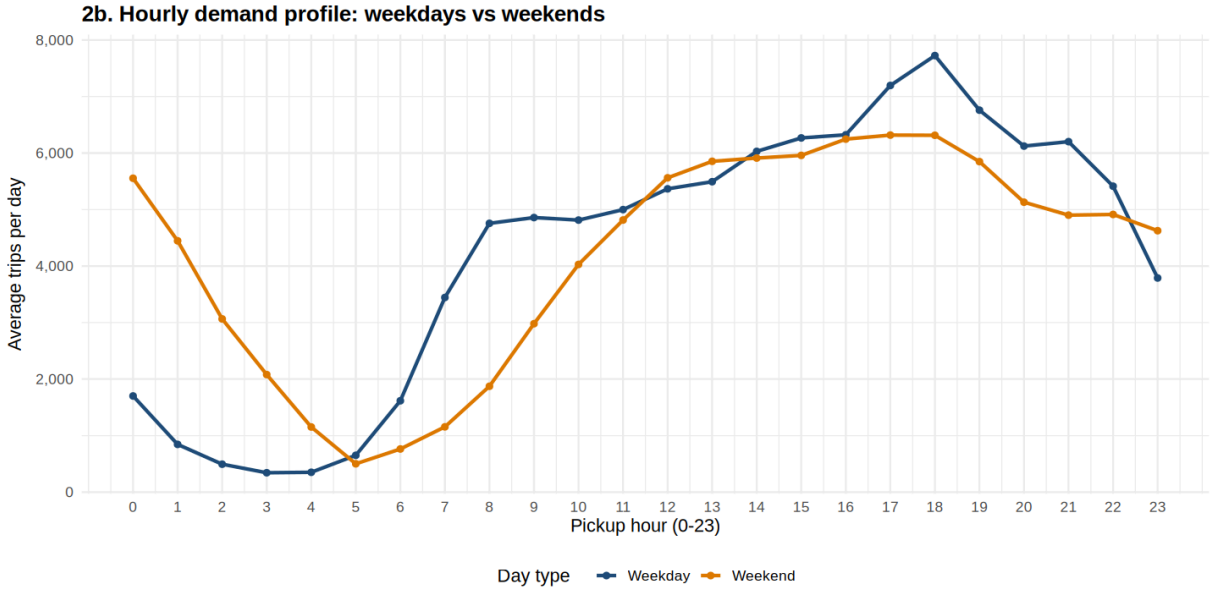
In [104…

```
p <- ggplot(dow, aes(pickup_wday, avg_trips_per_day / 1e3, fill =
pickup_wday %in% c("Sat", "Sun"))) +
  geom_col() +
  geom_text(aes(label = comma(avg_trips_per_day)), vjust = -0.4, size = 4)
+
  scale_fill_manual(values = c(`FALSE` = "#1F4E79", `TRUE` = "#F2B705"),
labels = c("Weekday", "Weekend"), name = NULL) +
  labs(title = "2. Taxi demand by day of week", subtitle = "Average number
of trips per calendar day",
       x = "Day of week", y = "Average trips per day (thousands)")
save_plot(p, "02_demand_by_weekday")
```



In [105…

```
wk <- melt(dcast(dow_hour, pickup_hour ~ ifelse(is_weekend, "Weekend",
"Weekday"), value.var = "trips_per_day"),
           id.vars = "pickup_hour", variable.name = "day_type", value.name
= "trips_per_day")
p <- ggplot(wk, aes(pickup_hour, trips_per_day, colour = day_type)) +
geom_line(linewidth = 1.2) + geom_point() +
  scale_colour_manual(values = c(Weekday = "#1F4E79", Weekend =
"#E07A00"), name = "Day type") +
  scale_x_continuous(breaks = 0:23) + scale_y_continuous(labels = comma) +
  labs(title = "2b. Hourly demand profile: weekdays vs weekends", x =
"Pickup hour (0-23)", y = "Average trips per day")
save_plot(p, "02b_weekday_vs_weekend_hourly")
```

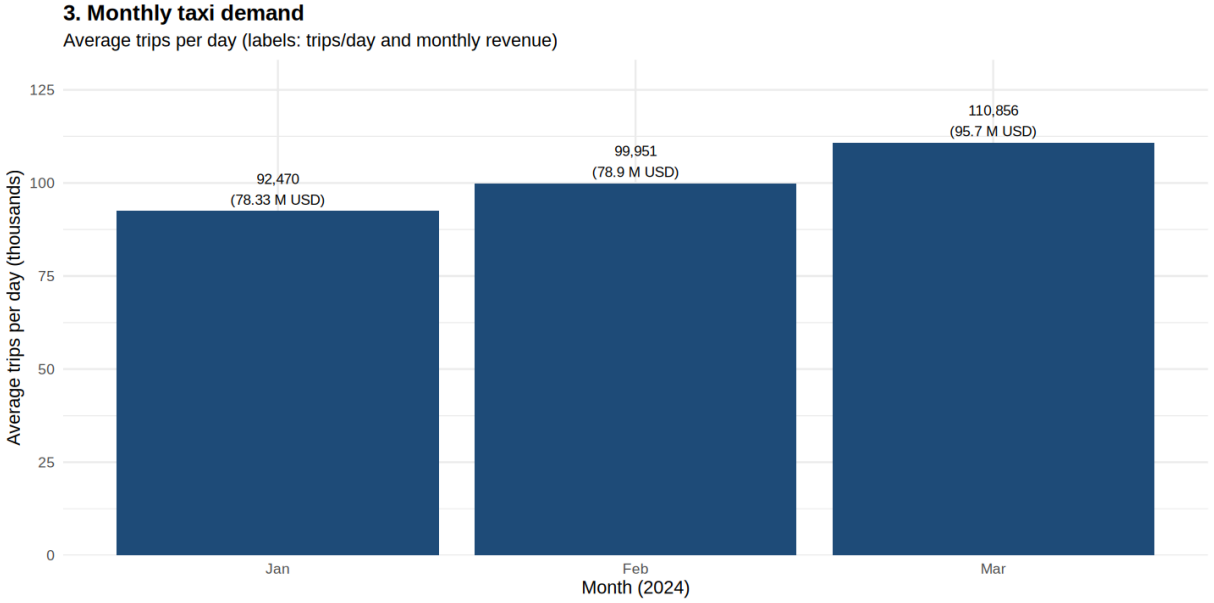


Weekdays show morning and evening commuter peaks; weekends have no morning peak but much higher late-night demand (0–3 h).

In [106…

```
p <- ggplot(monthly, aes(month, trips_per_day / 1e3)) +
  geom_col(fill = "#1F4E79") +
  geom_text(aes(label = paste0(comma(trips_per_day), "\n(", revenue_musd,
" M USD)")), vjust = -0.2, size = 3.6) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.2))) +
  labs(title = "3. Monthly taxi demand", subtitle = "Average trips per day
```

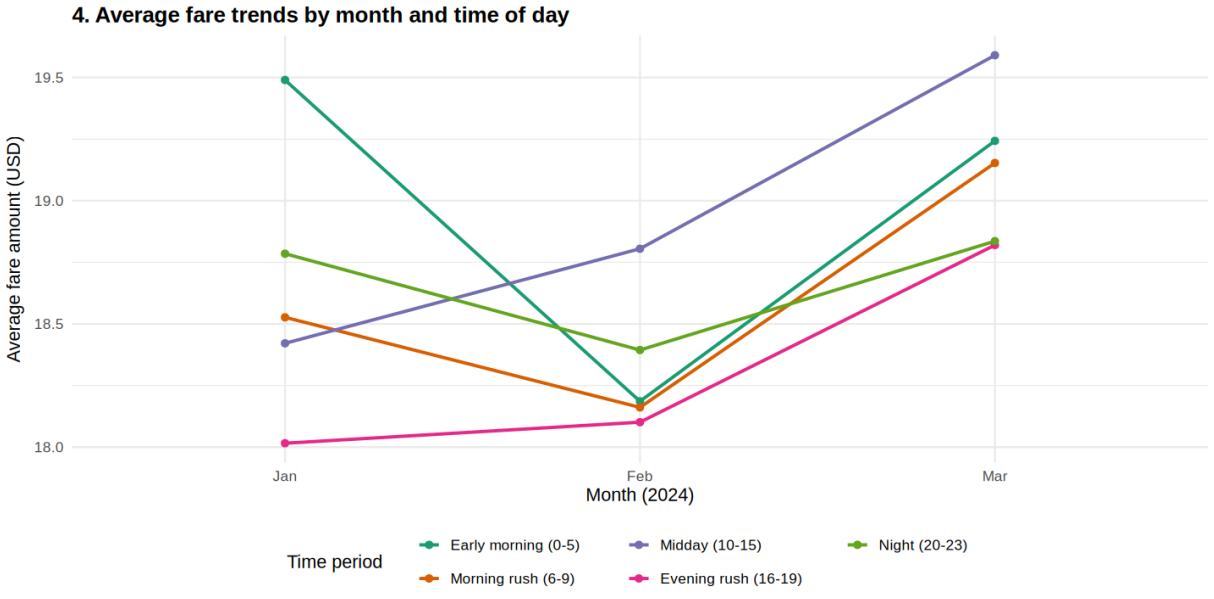
```
(labels: trips/day and monthly revenue)",
       x = "Month (2024)", y = "Average trips per day (thousands)")
save_plot(p, "03_monthly_demand")
```



January and February are the quietest months; demand peaks in spring (March–May).

In [107…

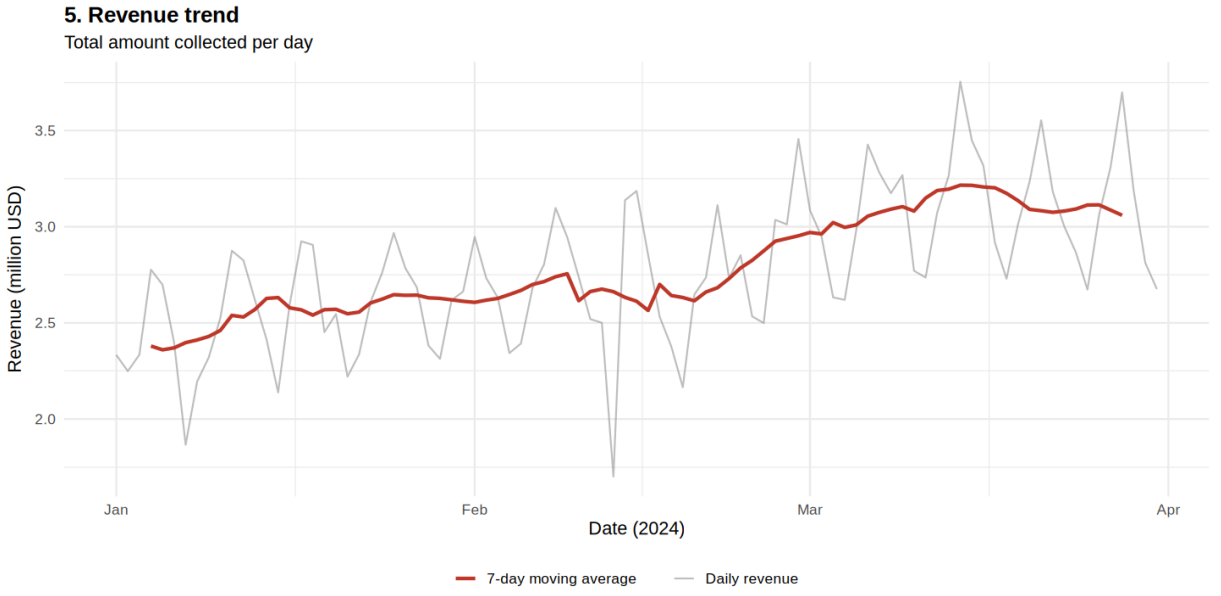
```
fare_trend <- trips[, .(avg_fare = mean(fare_amount), avg_total =
mean(total_amount)), by = .(month, time_period)]
p <- ggplot(fare_trend, aes(month, avg_fare, colour = time_period, group =
time_period)) +
  geom_line(linewidth = 1.1) + geom_point(size = 2) +
  scale_colour_brewer(palette = "Dark2", name = "Time period") +
  labs(title = "4. Average fare trends by month and time of day", x =
"Month (2024)", y = "Average fare amount (USD)") +
  guides(colour = guide_legend(nrow = 2))
save_plot(p, "04_average_fare_trends")
```



Early-morning trips are consistently the most expensive on average; fares are stable across months.

In [108…

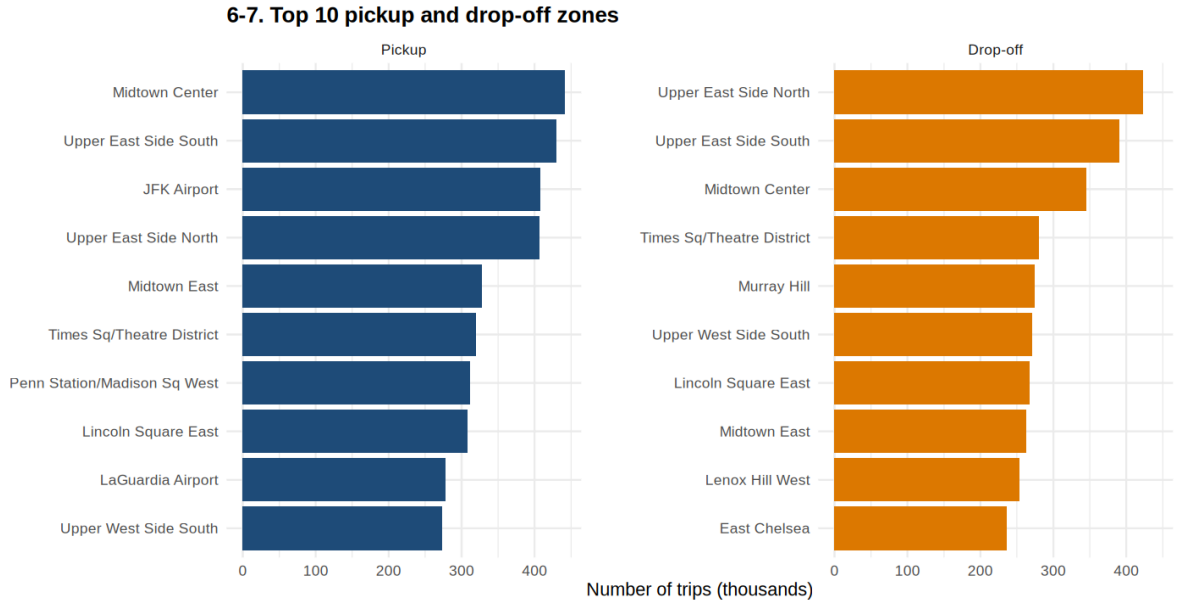
```
daily[, rev_7d := frollmean(revenue, 7, align = "center")]
p <- ggplot(daily, aes(as.Date(pickup_date))) +
  geom_line(aes(y = revenue / 1e6, colour = "Daily revenue"), alpha = 0.5)
+
  geom_line(aes(y = rev_7d / 1e6, colour = "7-day moving average"),
linewidth = 1.2, na.rm = TRUE) +
  scale_colour_manual(values = c(`Daily revenue` = "grey50", `7-day moving
average` = "#C0392B"), name = NULL) +
  scale_x_date(date_breaks = "1 month", date_labels = "%b") +
  labs(title = "5. Revenue trend", subtitle = "Total amount collected per
day", x = "Date (2024)", y = "Revenue (million USD)")
save_plot(p, "05_revenue_trend")
```



Revenue shows a strong weekly cycle and dips on holidays (e.g. New Year, Memorial Day) and stormy days.

In [109…

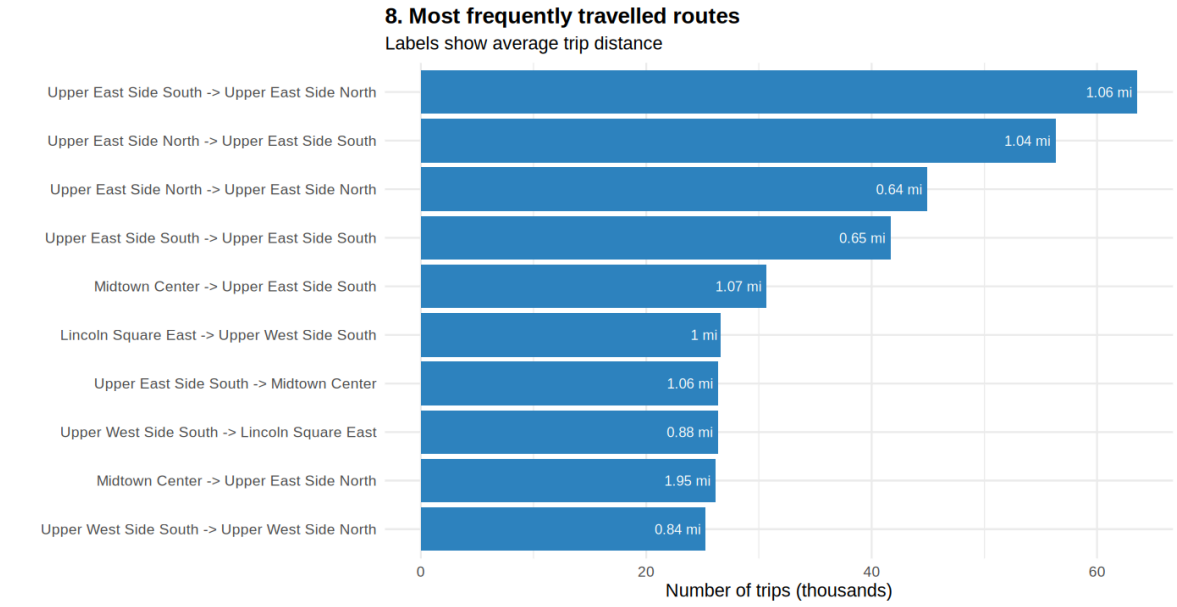
```
top_pu <- trips[, .N, by = pu_zone][order(-N)][1:10]
top_do <- trips[, .N, by = do_zone][order(-N)][1:10]
top_zones <- rbind(top_pu[, .(zone = pu_zone, N, type = "Pickup")],
top_do[, .(zone = do_zone, N, type = "Drop-off")])
top_zones[, zone_in_panel := reorder(paste(zone, type, sep = "___"), N)]
# sort bars separately in each panel
p <- ggplot(top_zones, aes(zone_in_panel, N / 1e3, fill = type)) +
geom_col(show.legend = FALSE) + coord_flip() +
  facet_wrap(~ factor(type, c("Pickup", "Drop-off")), scales = "free_y") +
  scale_x_discrete(labels = function(x) sub("___.*$", "", x)) +
  scale_fill_manual(values = c(Pickup = "#1F4E79", `Drop-off` =
"#E07A00")) +
  labs(title = "6-7. Top 10 pickup and drop-off zones", x = NULL, y =
"Number of trips (thousands)")
save_plot(p, "06_07_top_pickup_dropoff_zones", width = 13)
```



Midtown/Upper East Side and JFK dominate pickups; drop-offs are spread over residential Manhattan zones.

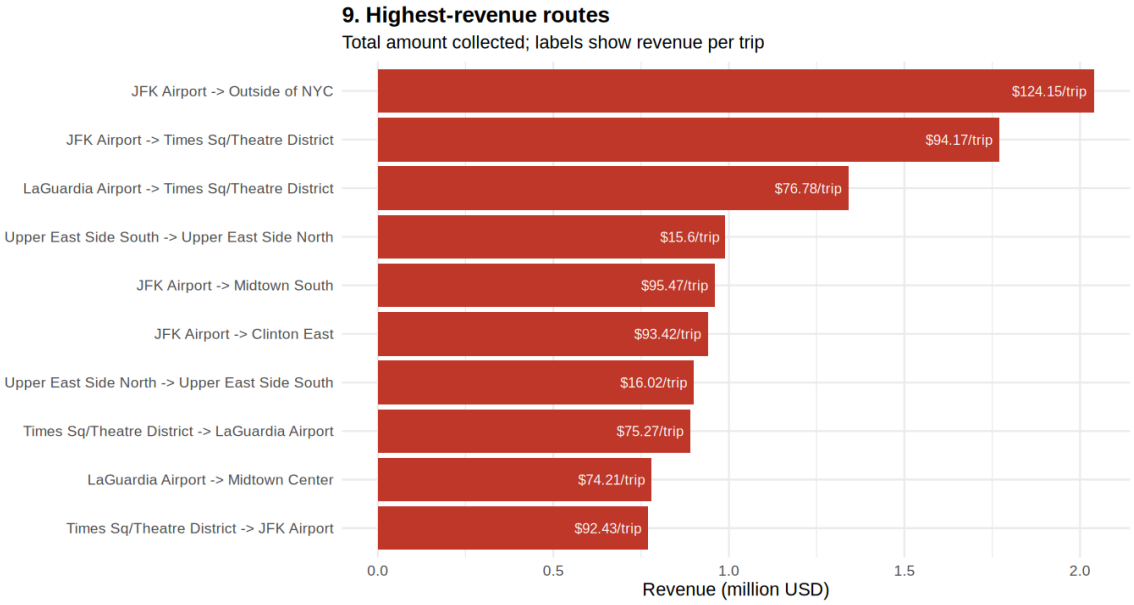
In [110…

```
tr <- copy(top_routes[1:10])[, route := paste(pu_zone, "->", do_zone)]
p <- ggplot(tr, aes(reorder(route, trips), trips / 1e3)) + geom_col(fill =
"#2E86C1") + coord_flip() +
  geom_text(aes(label = paste0(avg_distance, " mi")), hjust = 1.1, colour
= "white", size = 3.5) +
  labs(title = "8. Most frequently travelled routes", subtitle = "Labels
show average trip distance",
       x = NULL, y = "Number of trips (thousands)")
save_plot(p, "08_top_routes", width = 12)
```



In [111…

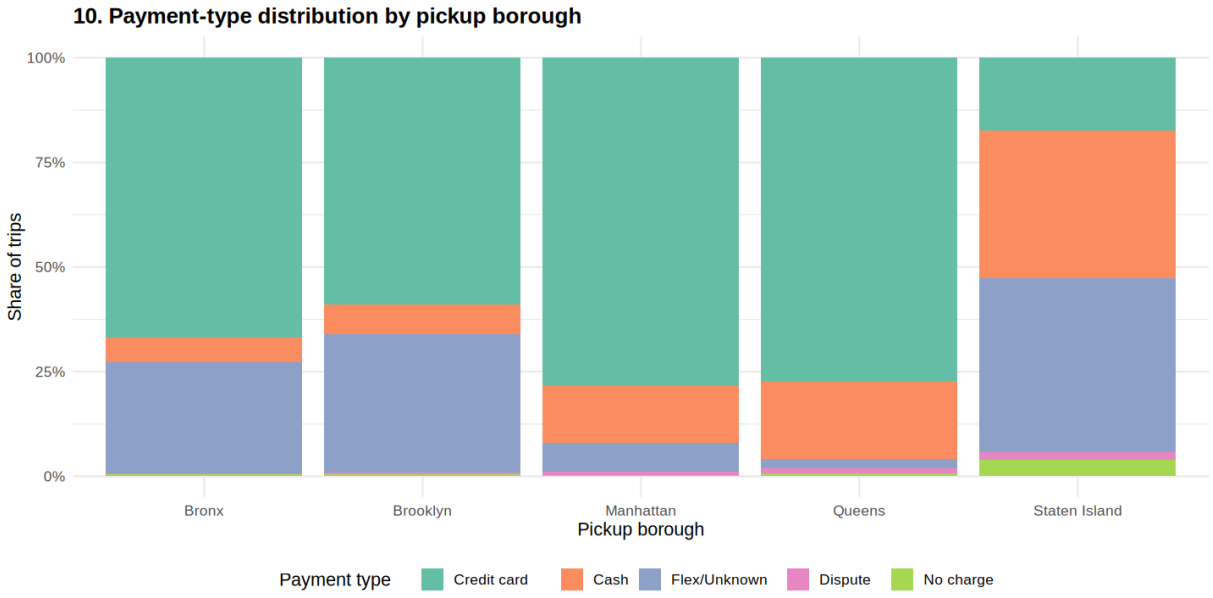
```
rr <- copy(top_rev_routes[1:10])[, route := paste(pu_zone, "->", do_zone)]
p <- ggplot(rr, aes(reorder(route, revenue_musd), revenue_musd)) +
geom_col(fill = "#C0392B") + coord_flip() +
  geom_text(aes(label = paste0("$", revenue_per_trip, "/trip")), hjust =
1.1, colour = "white", size = 3.5) +
  labs(title = "9. Highest-revenue routes", subtitle = "Total amount
collected; labels show revenue per trip",
       x = NULL, y = "Revenue (million USD)")
save_plot(p, "09_top_revenue_routes", width = 12)
```



Short Upper-East-Side hops are the most frequent routes, but the money is made on airport routes, which earn several times more per trip.

In [112…

```
pb <- pay_borough[pu_borough != "EWR"]
p <- ggplot(pb, aes(pu_borough, share_pct, fill = payment_type)) +
geom_col(position = "fill") +
  scale_y_continuous(labels = percent) + scale_fill_brewer(palette =
"Set2", name = "Payment type") +
  labs(title = "10. Payment-type distribution by pickup borough", x =
"Pickup borough", y = "Share of trips")
save_plot(p, "10_payment_by_borough")
```

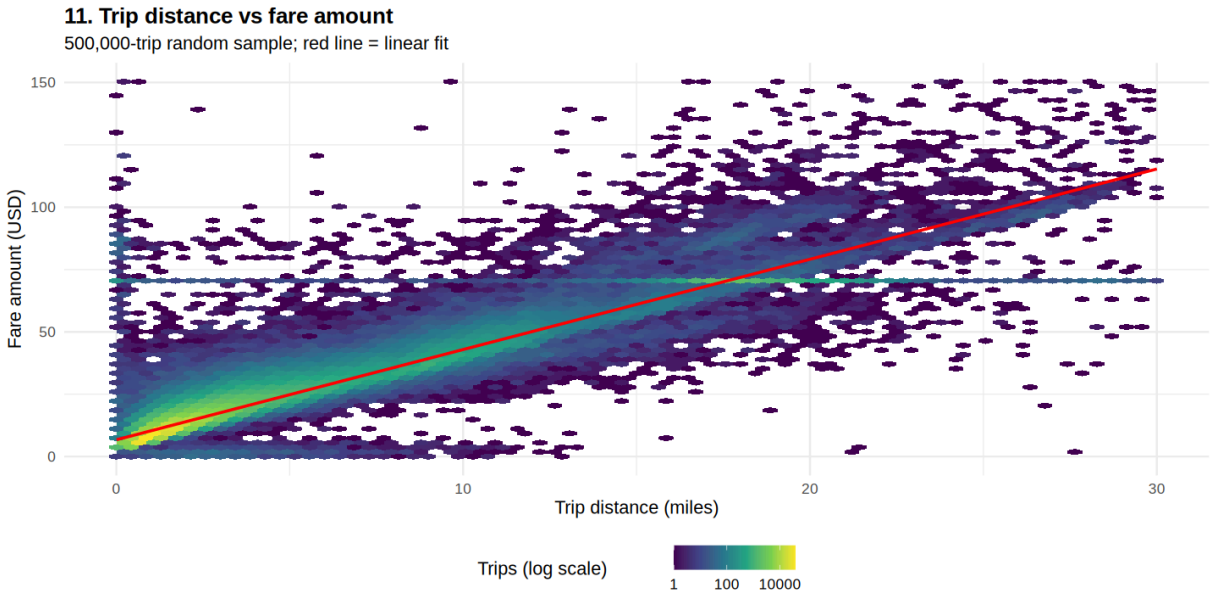


Credit cards dominate everywhere, but cash and flex/unknown payments are more common outside Manhattan.

In [113…

```
hex_samp <- trips[sample(.N, 5e5)][trip_distance <= 30 & fare_amount <=
150]
p <- ggplot(hex_samp, aes(trip_distance, fare_amount)) +
  geom_hex(bins = 70) + scale_fill_viridis_c(trans = "log10", name =
"Trips (log scale)") +
  geom_smooth(method = "lm", formula = y ~ x, colour = "red", linewidth =
```

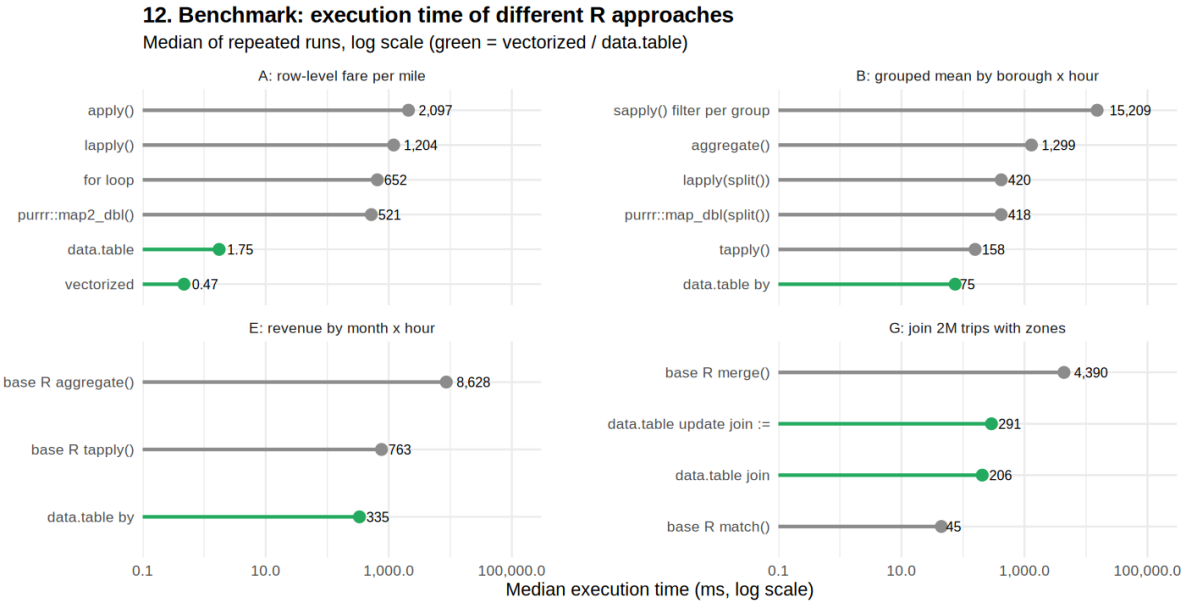
```
1) +
  labs(title = "11. Trip distance vs fare amount", subtitle = "500,000-
trip random sample; red line = linear fit",
       x = "Trip distance (miles)", y = "Fare amount (USD)")
save_plot(p, "11_distance_vs_fare")
```



Fare rises almost linearly with distance. The horizontal band at ≈ 70 USD is the JFK–Manhattan flat fare.

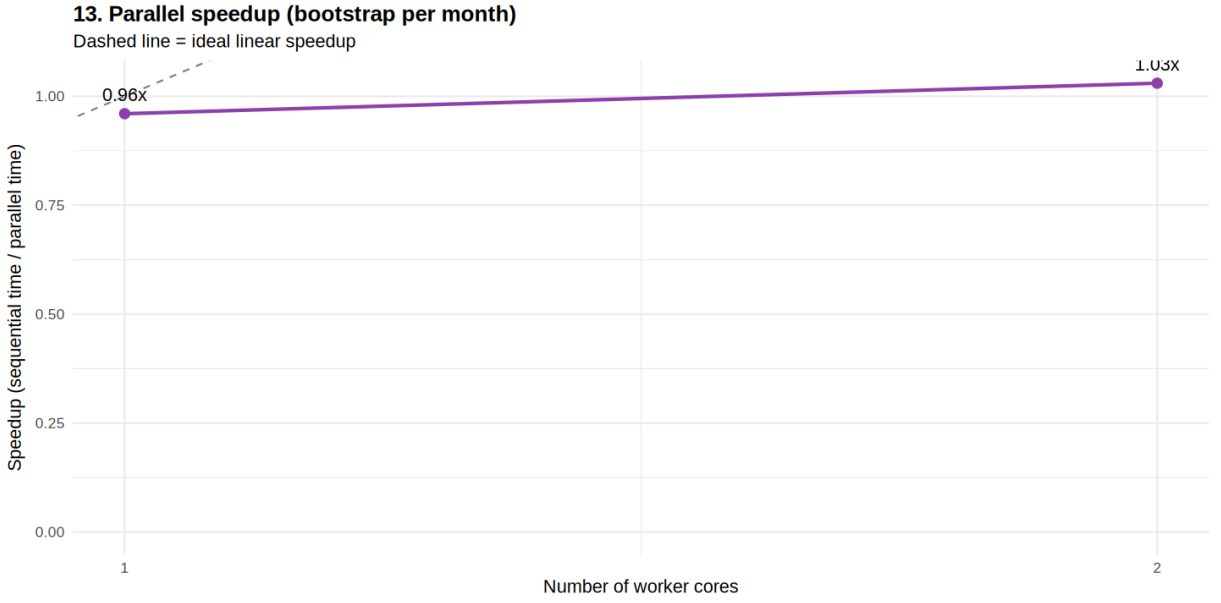
In [114…

```
bp <- bench_all[comparison %in% c("A: row-level fare per mile", "B:
grouped mean by borough x hour", "E: revenue by month x hour", "G: join 2M
trips with zones")]
bp[, fast_style := grepl("data.table|vectorized", approach)]
bp[, approach_in_panel := reorder(paste(approach, comparison, sep =
"___"), median_ms)]   # sort within each panel
p <- ggplot(bp, aes(approach_in_panel, median_ms, colour = fast_style)) +
  geom_segment(aes(xend = approach_in_panel, y = 0.1, yend = median_ms),
linewidth = 1.2, show.legend = FALSE) +
  geom_point(size = 3.5, show.legend = FALSE) +
  geom_text(aes(label = ifelse(median_ms >= 10, comma(median_ms, accuracy
= 1), round(median_ms, 2))), hjust = -0.3, size = 3.4, colour = "black") +
  coord_flip() + facet_wrap(~ comparison, scales = "free_y", ncol = 2) +
  scale_x_discrete(labels = function(x) sub("___.*$", "", x)) +
  scale_y_log10(labels = comma, expand = expansion(mult = c(0, 0.25))) +
  scale_colour_manual(values = c(`FALSE` = "grey55", `TRUE` = "#27AE60"))
+
  labs(title = "12. Benchmark: execution time of different R approaches",
subtitle = "Median of repeated runs, log scale (green = vectorized /
data.table)",
       x = NULL, y = "Median execution time (ms, log scale)")
save_plot(p, "12_benchmark_comparison", width = 14, height = 8)
```



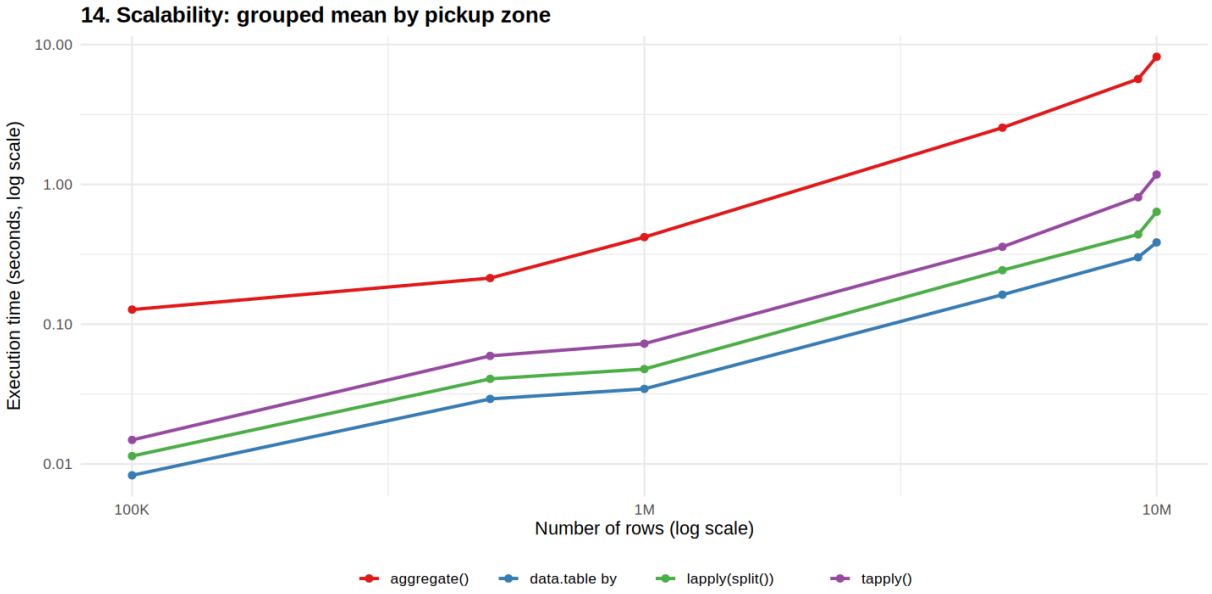
In [115…

```
p <- ggplot(scaling, aes(cores, speedup)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", colour =
"grey50") +
  geom_line(colour = "#8E44AD", linewidth = 1.2) + geom_point(size = 3,
colour = "#8E44AD") +
  geom_text(aes(label = paste0(speedup, "x")), vjust = -1) +
  scale_x_continuous(breaks = scaling$cores) + expand_limits(y = 0) +
  labs(title = "13. Parallel speedup (bootstrap per month)", subtitle =
"Dashed line = ideal linear speedup",
       x = "Number of worker cores", y = "Speedup (sequential time /
parallel time)")
save_plot(p, "13_parallel_speedup", width = 9)
```



In [116…

```
p <- ggplot(scal, aes(rows, median_ms / 1000, colour = approach)) +
geom_line(linewidth = 1.1) + geom_point(size = 2) +
  scale_x_log10(labels = label_number(scale_cut = cut_short_scale())) +
scale_y_log10() +
  scale_colour_brewer(palette = "Set1", name = NULL) +
  labs(title = "14. Scalability: grouped mean by pickup zone", x = "Number
of rows (log scale)", y = "Execution time (seconds, log scale)")
save_plot(p, "14_scalability")
```



data.table is fastest for every size and the gap widens as the number of rows grows; parallel speedup is real but sub-linear.

8.1 Optional advanced task – geographical visualizations

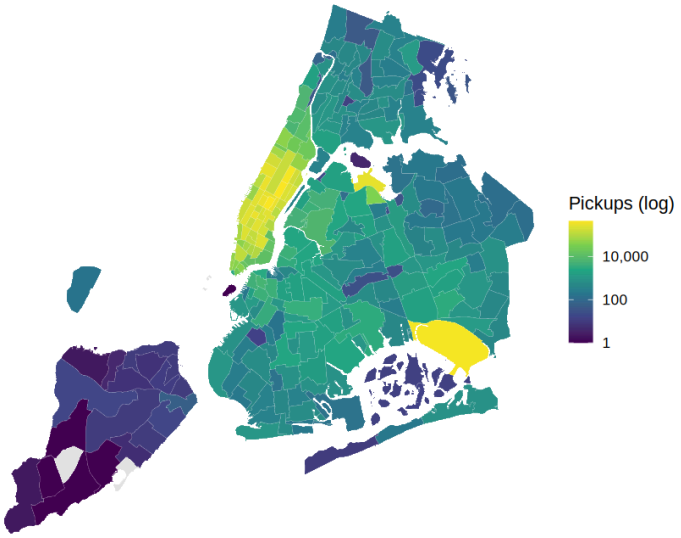
Zone polygons from the official TLC shapefile are used for a static choropleth (ggplot2 + sf) and an interactive **leaflet** map of the top pickup zones.

In [117…

```
if (!HAVE_MAPS) {
  message("sf / leaflet are not available - skipping the optional map
section.")
} else {
suppressPackageStartupMessages({ library(sf); library(leaflet);
library(htmlwidgets) })
shape_dir <- file.path(DATA_DIR, "taxi_zones")
if (!dir.exists(shape_dir)) unzip(shape_zip, exdir = shape_dir)
shp <- list.files(shape_dir, pattern = "\\.shp$", recursive = TRUE,
full.names = TRUE)[1]
zones_sf <- st_transform(st_make_valid(st_read(shp, quiet = TRUE)), 4326)
zone_stats <- trips[, .(pickups = .N, revenue = sum(total_amount),
avg_total = mean(total_amount)), by = .(LocationID = PULocationID)]
zones_sf <- merge(zones_sf, zone_stats, by = "LocationID", all.x = TRUE)

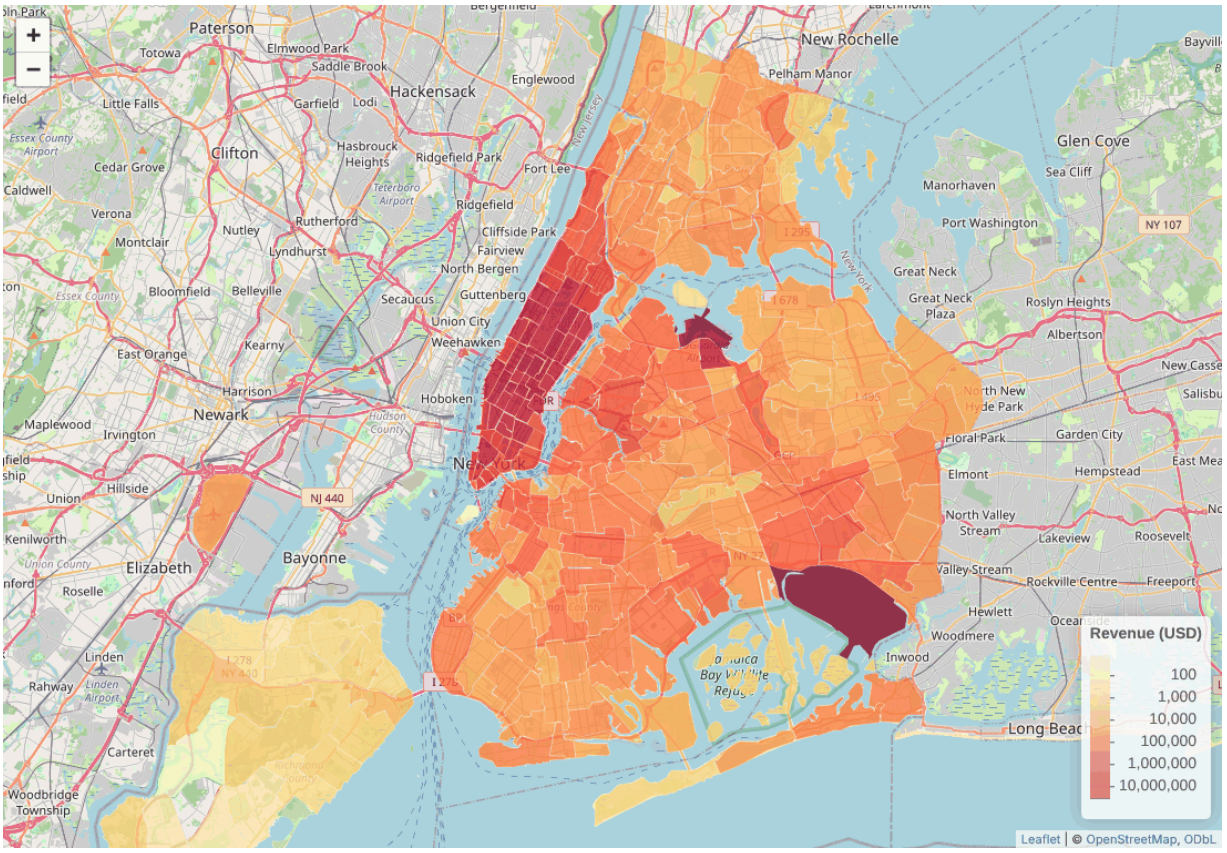
p <- ggplot(zones_sf) + geom_sf(aes(fill = pickups), colour = "white",
linewidth = 0.05) +
  scale_fill_viridis_c(trans = "log10", labels = comma, name = "Pickups
(log)", na.value = "grey90") +
  labs(title = paste("15. Pickups by taxi zone,", PERIOD_LABEL)) +
  theme_void(base_size = 13) + theme(legend.position = "right", plot.title
= element_text(face = "bold"))
save_plot(p, "15_pickup_map", width = 9, height = 9)
}
```

15. Pickups by taxi zone, Jan-Mar 2024



In [118…

```
if (HAVE_MAPS) {
# an interactive choropleth: zone polygons never overlap, unlike circle
markers
map_sf <- zones_sf[!is.na(zones_sf$revenue), ]
pal <- colorNumeric("YlOrRd", domain = log10(map_sf$revenue))
m <- leaflet(map_sf) |>
  addTiles() |>                                   # plain OpenStreetMap
tiles, no API key needed
  setView(lng = -73.93, lat = 40.73, zoom = 11) |>
  addPolygons(fillColor = ~pal(log10(revenue)), fillOpacity = 0.75, weight
= 0.6, color = "white",
              highlightOptions = highlightOptions(weight = 2.5, color =
"#1F4E79", bringToFront = TRUE),
              label = ~sprintf("%s (%s): %s pickups, $%s M revenue, $%.2f
per trip",
                               zone, borough, comma(pickups),
round(revenue / 1e6, 2), avg_total)) |>
  addLegend("bottomright", pal = pal, values = ~log10(revenue), title =
"Revenue (USD)",
            labFormat = labelFormat(transform = function(x) round(10^x)))
saveWidget(m, file.path(normalizePath(OUT_DIR),
"leaflet_revenue_by_zone.html"), selfcontained = TRUE)
m
}
```



*Each taxi zone is shaded by the **total revenue collected from trips starting there** (log scale). Midtown, the Upper East Side and the two airports are the dark, high-revenue areas; hover over any zone to see its pickups, revenue and average amount per trip.*

> **Note on the saved notebook:** an interactive leaflet widget embeds the whole map library (about 4 MB), which stops GitHub from rendering the notebook in the browser. The committed notebook therefore shows a static preview of this map. Run the cell yourself, or open `outputs/leaflet_revenue_by_zone.html`, for the fully interactive version.

---

9. Observations

The key measured numbers from this run are collected below (they will vary slightly between machines).

In [119…

```
bm <- function(cmp, appr) bench_all[comparison == cmp & approach == appr,
median_ms]
obs <- data.table(
  metric = c("Trips analysed (after cleaning)", "Peak pickup hour",
"Busiest weekday (avg trips/day)", "Busiest month (avg trips/day)",
             "Top revenue route", "Credit-card share of trips (%)",
"Distance-fare correlation",
             "Row-level: apply() vs vectorized (x slower)", "Grouping 2M:
lapply(split()) vs data.table (x slower)",
             sprintf("Grouping %s rows: aggregate() vs data.table (x
slower)", format(nrow(trips), big.mark = ",")),
```

```
            "Join 2M: merge() vs data.table (x slower)",
            sprintf("Parallel speedup with %d cores", N_CORES), "Light
task: parallel vs sequential (speedup)"),
  value = c(format(nrow(trips), big.mark = ","),
paste0(hourly[which.max(trips), pickup_hour], ":00"),
          as.character(dow[which.max(avg_trips_per_day), pickup_wday]),
as.character(monthly[which.max(trips_per_day), month]),
          top_rev_routes[1, paste(pu_zone, "->", do_zone)],
pay_overall[payment_type == "Credit card", share_pct],
          round(trips[, cor(trip_distance, fare_amount)], 3),
          round(bm("A: row-level fare per mile", "apply()") / bm("A:
row-level fare per mile", "vectorized")),
          round(bm("B: grouped mean by borough x hour",
"lapply(split())") / bm("B: grouped mean by borough x hour", "data.table
by")),
          round(bm("E: revenue by month x hour", "base R aggregate()") /
bm("E: revenue by month x hour", "data.table by")),
          round(bm("G: join 2M trips with zones", "base R merge()") /
bm("G: join 2M trips with zones", "data.table join")),
          paste0(speedup_main, "x"), paste0(light[2, speedup_vs_seq],
"x")))
obs
```

A data.table: 13 × 2

metric	value
<chr>	<chr>
Trips analysed (after cleaning)	9,201,683
Peak pickup hour	18:00
Busiest weekday (avg trips/day)	Thu
Busiest month (avg trips/day)	Mar
Top revenue route	JFK Airport → Outside of NYC
Credit-card share of trips (%)	78.14
Distance-fare correlation	0.942
Row-level: apply() vs vectorized (x slower)	4461
Grouping 2M: lapply(split()) vs data.table (x slower)	6
Grouping 9,201,683 rows: aggregate() vs data.table (x slower)	26
Join 2M: merge() vs data.table (x slower)	21
Parallel speedup with 2 cores	1x
Light task: parallel vs sequential (speedup)	0.01x

1. **Demand** is strongly time dependent: an evening peak around 17–18 h, a minimum at 4–5 AM, the highest per-day demand on mid-week days and the lowest on Sundays, and a spring high after the winter months.
2. **Revenue** is concentrated in the midday and evening-rush periods, while **airport routes** (JFK, LaGuardia) generate by far the highest revenue per trip and

dominate the highest-revenue routes.
3. **Fare vs distance** is strongly linear, but the fare per mile is highest for very short trips.
4. **Payments:** credit cards account for about three quarters of trips; cash and unrecorded ("flex/unknown") payments are relatively more common outside Manhattan, and tips are recorded almost only for card payments.
5. **Performance:** data.table and vectorized operations were faster than functional row-wise code by orders of magnitude; data.table beat base R `aggregate()` / `merge()` by a large factor while using less memory; parallel processing helped the heavy bootstrap only when there were enough cores and partitions to keep the workers busy, and was slower than sequential for the lightweight task on every machine.

---

10. Task 7 – Critical Analysis and Recommendation

1. Which approach provides the best execution performance? data.table for data manipulation (filter, group, sort, join) and **vectorized** code for element-wise arithmetic. In the benchmarks data.table was the fastest entry for filtering, grouping and sorting, many times faster than base R `aggregate()` and `merge()`, and vectorized arithmetic was thousands of times faster than row-wise `apply()`. The exceptions are instructive: a single-column `match()` lookup beat the data.table join, and functional `lapply()` over a few columns matched data.table – iterating over a few large units costs nothing.

2. How does data.table improve the processing of large datasets? (a) Operations are implemented in optimised, **multi-threaded C** (radix sorting and grouping, GForce for `mean` / `sum`); (b) `:=`, `set()`, `setkey()`, `setorder()` and update joins modify data **by reference**, so a 20-million-row table is never copied – visible in the much lower peak memory of the update join; (c) `fread()` / `setDT()` give fast import; (d) the concise `DT[i, j, by]` syntax combines filtering, computing and grouping in one optimised step.

3. When is functional programming preferable? When iterating over a **small number of large items** – columns, files, months, model specifications, groups – where the per-call overhead is negligible (Experiment C showed `lapply()` / `map_dbl()` as fast as `colMeans()`), and when clarity and composability matter (purrr's type-stable `map_dbl()`, pipelines, applying the same function to every partition). It is also the natural interface to parallelism (`foreach`, `parLapply`). It should not be used to loop over millions of rows.

4. When do vectorized operations provide better performance? Whenever the same operation is applied to every element of a vector/column (arithmetic, comparisons, `ifelse`, `pmin`, cumulative functions): one call into compiled code replaces millions of interpreted R calls, giving the 1000×-scale gains measured in Experiment A.

5. What performance improvement was obtained using parallel computing? The measured figure is in the scaling table (Section 6.5), in the interpretation printed in Section 6.7 and in the observations table. On a 4-core machine with six monthly partitions the bootstrap ran about **2× faster** on 3 workers (efficiency roughly 70–90 %). On a 2-core Colab session with three partitions there is **no gain at all**: three partitions over two workers allow at most 1.5×, and the cost of starting the workers and copying the partitions to them cancels even that. The gain is therefore a property of the machine and of the partitioning, not of `foreach` itself.

6. Does parallel computing always result in faster execution? No, and this notebook measured both outcomes. The lightweight task (mean fare per month) was tens of times slower in parallel, because starting the workers and serialising the partitions costs far more than the arithmetic; a single multi-threaded data.table grouping beat both partitioned versions. The heavy bootstrap gained about 2× on 4 cores with six partitions, but **nothing** on a 2-core session with three partitions. Three causes explain that: too few partitions for the number of workers (3 over 2 workers caps the speedup at 1.5×), the cost of copying every partition into a separate R process, and hyperthreaded vCPUs that are not two independent cores. Parallelism helps only when the work per partition is large relative to the communication cost **and** there are enough partitions to keep every worker busy.

7. Trade-offs between speed, memory, readability and scalability

Approach	Speed	Memory	Readability	Scalability
`apply()` / `lapply()` / `map()` per row	very slow	high (per-call allocation, matrix copy)	good	poor
`for` loop (pre-allocated)	very slow	low	good for beginners, verbose	poor
Vectorized base R	very fast	moderate (temporary vectors)	very good	good
Base R `aggregate()` / `merge()`	slow	high (copies)	good	moderate
data.table	fastest	lowest (by reference)	good once learnt, terse syntax	excellent
foreach + doParallel	faster only for heavy tasks	high (data copied to every worker)	more complex code	good for CPU-bound, independent tasks

8. Recommendation for large-scale analytical applications Use **data.table as the core engine** (import with arrow/fread, clean, join and aggregate by reference), write element-wise logic in **vectorized** form, use **functional programming** (`lapply` / `purrr::map`) for iterating over files, columns or partitions, and add **foreach + doParallel** only for genuinely CPU-intensive, independent tasks such as

bootstrapping, simulation or model fitting per partition – after checking with a benchmark that the speedup outweighs the overhead. This recommendation is directly supported by the benchmark and speedup tables obtained in this notebook.

---

11. Conclusion

An end-to-end, reproducible big-data workflow was built in R for ≈ 20 million NYC Yellow Taxi trips (January–June 2024). data.table made it possible to import, clean, enrich, join and aggregate the full dataset in seconds on a laptop with modest memory, by working in place. The analysis showed clear temporal demand patterns, the dominance of credit-card payments and the outsized revenue of airport routes. The benchmarks demonstrated that **how** R code is written matters more than the hardware: row-wise functional code was thousands of times slower than vectorized code, base R grouping and joining were many times slower and more memory hungry than data.table, and parallel computing with foreach/doParallel delivered a useful but sub-linear speedup only for compute-heavy work. The recommended architecture is therefore data.table + vectorization at the core, functional programming for iteration over a few large units, and parallelism only where benchmarks prove it worthwhile.

In [120…

```
sessionInfo()
```

```
R version 4.6.1 (2026-06-24)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.5 LTS

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;  LAPACK version 3.12.0

locale:
 [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C               LC_TIME=en_US.UTF-8
LC_COLLATE=en_US.UTF-8     LC_MONETARY=en_US.UTF-8
 [6] LC_MESSAGES=en_US.UTF-8    LC_PAPER=en_US.UTF-8       LC_NAME=C                  LC_ADDRESS=C
LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: Etc/UTC
tzcode source: system (glibc)

attached base packages:
[1] parallel  stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] htmlwidgets_1.6.4    leaflet_2.2.3        sf_1.1-3             scales_1.4.0         lubridate_1.9.5
purrr_1.2.2
 [7] microbenchmark_1.5.0 doParallel_1.0.17    iterators_1.0.14     foreach_1.5.2        ggplot2_4.0.3
arrow_26.0.0
[13] data.table_1.18.6.1

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       xfun_0.62          lattice_0.23-1     vctrs_0.7.3        tools_4.6.1
crosstalk_1.2.2    generics_0.1.4
 [8] tibble_3.3.1       proxy_0.4-29       pkgconfig_2.0.3    Matrix_1.7-6       KernSmooth_2.23-27
RColorBrewer_1.1-3 S7_0.2.2
[15] assertthat_0.2.1   uuid_1.2-2         lifecycle_1.0.5    compiler_4.6.1     farver_2.1.2
textshaping_1.0.5  repr_1.1.7
[22] codetools_0.2-20   htmltools_0.5.9    class_7.3-24       yaml_2.3.12        pillar_1.11.1
hexbin_1.28.6      crayon_1.5.3
[29] jquerylib_0.1.4    classInt_0.4-11    nlme_3.1-171       tidyselect_1.2.1   digest_0.6.39
dplyr_1.2.1        labeling_0.4.3
[36] splines_4.6.1      fastmap_1.2.0      grid_4.6.1         cli_3.6.6          magrittr_2.0.5
base64enc_0.1-6    IRdisplay_1.1
[43] e1071_1.7-17       withr_3.0.3        IRkernel_1.3.2     bit64_4.8.6        timechange_0.4.0
rmarkdown_2.32     bit_4.6.0
[50] otel_0.2.0         pbdZMQ_0.3-14      ragg_1.5.2         evaluate_1.0.5     knitr_1.52
viridisLite_0.4.3  mgcv_1.9-4
[57] rlang_1.3.0        Rcpp_1.1.2         glue_1.8.1         DBI_1.3.0          jsonlite_2.0.0
R6_2.6.1           systemfonts_1.3.2
[64] units_1.0-1
```